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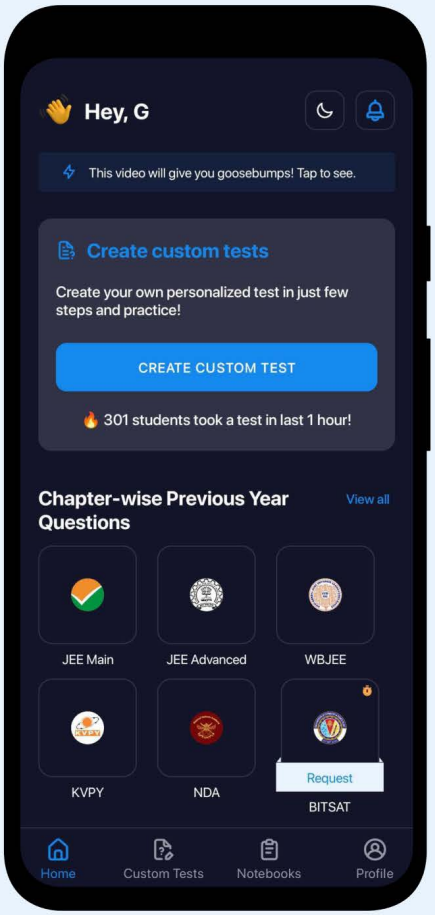

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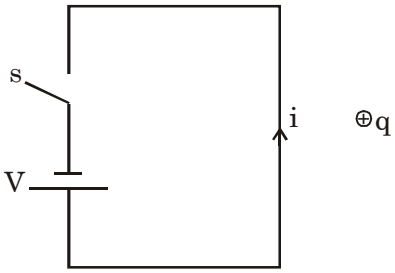
MAGNETISM

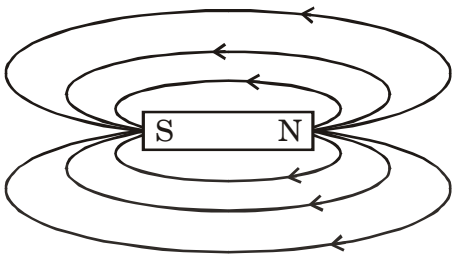
Quizrr

1. Concept of Magnetic Field :

What is magnetic field and how is it Generated :

Consider the situation when a charge placed near an electrical circuit. When the switch is closed and the charge is at rest, then there is no force between the current carrying wire and charged particle. Now, it was experimentally observed that when the charge is set into motion, we observe a force acting on the charge. This suggests that some field is generated by the motion of a charge. This is known as **magnetic field.**





A magnet

Also, in case of permanent magnet. We have permanent magnetic field because the electrons present inside the atoms rotate in the same sense.

2. Motion of a Charge Inside Magnetic Field :

When a charge is moving a magnetic field, then the force acting on that charge is given by

$$\boxed{\vec{F}_m = q(\vec{v} \times \vec{B})} \quad \text{... (i)}$$

This is called **Lorentz force.** Here. $\vec{F}_m$ is the magnetic force on a charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$.

Following points are worth noting regarding the above expressions.

(i) The magnitude of $\vec{F}_m$ is,

$$\boxed{F_m = Bqv \sin\theta}$$

where θ is the angle between $\vec{v}$ and $\vec{B}$.

(ii) $\vec{F}_m$ is zero when,

(a) B = 0, i.e., no magnetic field is present.

(b) q = 0, i.e., particle is neutral.

(c) v = 0, i.e., charged particle is at rest or

(d) $\theta = 0$ or 180 , i.e., $\vec{v} \uparrow\uparrow \vec{B}$ or $\vec{v} \uparrow\downarrow \vec{B}$

(iii) $\vec{F}_m$ is maximum at $\theta = 90$ and this maximum value is Bqv.

(iv) The units of $\vec{B}$ must be the same as the units of F/ qv. Therefore, the SI unit of $\vec{B}$ is equivalent to $\frac{N-s}{C-m}$. This unit is called the **tesla** (abbreviated as T), in honour of Nikola Tesla, the prominent Serbian-American scientist and inventor. Thus,

$$1\,\text{tesla} = 1T = \frac{1N\text{-}s}{C\text{-}m} = \frac{1N}{A-m}$$

The CGS unit of $\vec{B}$, the **gauss** ($1G = 10^{-4}$ T) is also in common use.

(v) The formula is true for both uniform and non-uniform magnetic field.

(vi) In equation number (i) q is to be substituted with sign. If q is positive magnetic force is along $\vec{v} \times \vec{B}$ and if q is negative magnetic force is in a direction opposite to $\vec{v} \times \vec{B}$.

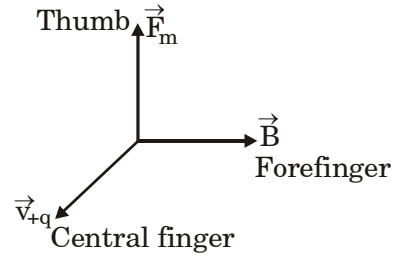
(vii) Direction of $\vec{F}_m$: From the property of cross product we can infer that $\vec{F}_m$ is perpendicular to both $\vec{v}$ and $\vec{B}$ or it is perpendicular to the plane formed by $\vec{v}$ and $\vec{B}$. The exact direction of $\vec{F}_m$ can be given by any of the following methods:

(a) Direction of $\vec{F}_m$ = (sign of q) (direction of $\vec{v} \times \vec{B}$) or, as we stated earlier also,

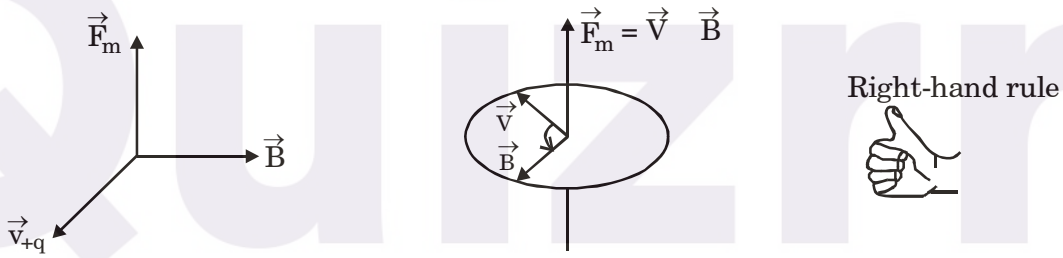
$$\vec{F}_m \uparrow\uparrow \vec{v} \quad \vec{B} \text{ if q is positive and}$$

$$\vec{F}_m \uparrow\downarrow \vec{v} \quad \vec{B} \text{ if q is negative.}$$

(b) **Fleming's left hand rule :** According to this rule, the forefinger, the central finger and the thumb of the left hand are stretched in such a way that they are mutually perpendicular to each other. If the central finger shows the direction of velocity of positive charge ($\vec{v}_{+q}$) and forefinger shows the direction of magnetic field ($\vec{B}$), then the thumb will give the direction of magnetic force $\vec{F}_m$. If instead of positive charge we have the negative charge, then is in opposite direction.



(c) **Right hand rule :** Wrap the fingers of your right hand around the line perpendicular to the plane of $\vec{v}$ and $\vec{B}$ as shown in figure, so that they curl around with the sense of rotation from $\vec{v}$ and $\vec{B}$ through the smaller angle between them. Your thumb then points in the direction of the force $\vec{F}_m$ on a positive charge. (Alternatively, the direction of the force $\vec{F}_m$ on a positive charge is the direction in which a right hand thread screw would advance if turned the same way).

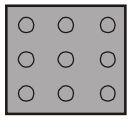


(viii) $\vec{F}_m \perp \vec{v}$ or $\frac{\vec{ds}}{dt}$. Therefore, $\vec{F}_m \perp \vec{ds}$ or the work done by the magnetic force in a static magnetic field is zero.

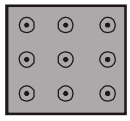
$$\boxed{W_{\vec{F}_m} = 0}$$

So, from work energy theorem KE and hence the speed of the charged particle remains constant in magnetic field. The magnetic force can change the direction only. It can not increase or decrease the speed or kinetic energy of the particle.

Note : By convention the direction of magnetic field B perpendicular to the paper going inwards is shown by ◯ and the direction perpendicular to the paper coming out is shown by ⊙.



Direction of $\vec{B}$ perpendicular to paper inwards



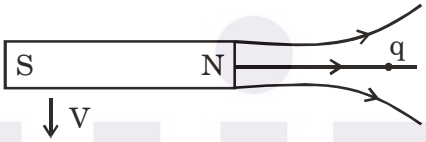
Direction of $\vec{B}$ perpendicular to paper outwards

Illustration :

Can a magnet set a resting electron into motion ?

Solution :

Yes. For example, the charge is at rest but the magnet is moved as shown in the figure, then the charge will be set into motion. this is because in this situation relative to the magnet, the charge is moving with a velocity v and hence it will experience a force perpendicular to the plane containing the field and motion of magnet.



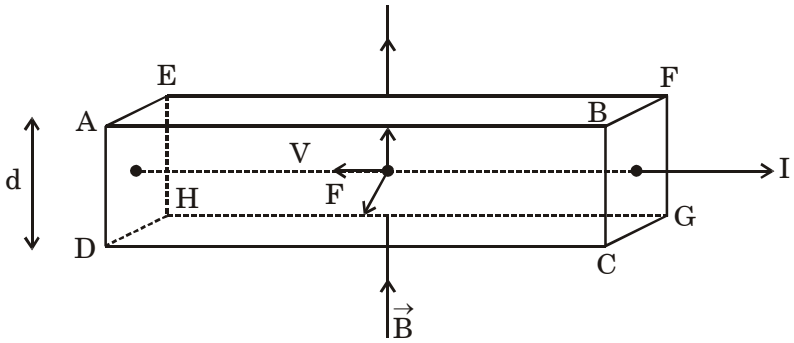
Note : Actually in this situation the field is not steady, and a time varying magnetic field produces an electric field due to which charge experiences a force and sets in motion.

Illustration :

A metallic block carrying current I is subjected to a uniform magnetic field $\vec{B}$ as shown. Which face will acquire lower potential if electrons inside the block move with velocity v **?**

Solution :

As the block is of metal, the charge carriers are electrons; so for current along positive x – axis, the electrons are moving along negative x-axis, i.e., $\vec{v} = -v\tilde{i}$



and as the magnetic field is along the y-axis, i.e. 1 $\vec{B} = B\tilde{j}$,

So, $$\vec{F} = q(\vec{v} \times \vec{B})$$

$$\vec{F} = -e(-v\tilde{i} \times B\tilde{j})$$

i.e. $$\vec{F} = evB\tilde{k}$$

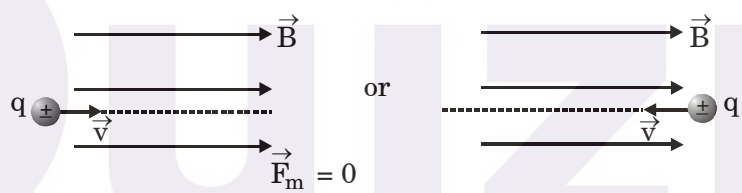
As the force on electrons is towards face ABCD, the electrons will accumulate on it and hence it will acquire lower potential.

3. Path of A Charged Particle in Uniform Magnetic Field

The path of a charged particle in uniform magnetic field depends on the angle θ (the angle between $\vec{v}$ and $\vec{B}$). Depending on the different values of θ, following three cases are possible.

Case 1. When θ is 0 or 180

$\vec{F}_m = 0$, when θ is either 0 or 180 . Hence, path of the charged particle is a straight line (undeviated) when it enters parallel or antiparallel to magnetic field.



Case 2. When θ = 90

When θ = 90 , the magnetic force is $F_m = Bqv \sin 90 = Bqv$. This magnetic force is perpendicular to the velocity at every instant. Hence, path is a **circle**. The necessary centripetal force is provided by the magnetic force. Hence, if r be the radius of the circle, then

$$\frac{mv^2}{r} = Bqv$$

or $$r = \frac{mv}{Bq}$$

This expression of r can be written in following different ways :

$$\boxed{r = \frac{mv}{Bq} = \frac{P}{Bq} = \frac{\sqrt{2Km}}{Bq} = \frac{\sqrt{2qVm}}{Bq}}$$

Here, P = momentum of particle

$$K = \text{K.E. of particle} = \frac{P^2}{2m} \quad \text{or} \quad P = \sqrt{2Km}$$

We also know that if the charged particle is accelerated by a potential difference of V volts, it acquires a KE given by,

$$K = qV$$

Further, time period of the circular path will be

$$T = \frac{2\pi r}{v} = \frac{2\pi\left(\frac{mv}{Bq}\right)}{v} = \frac{2\pi m}{Bq}$$

or

$$\boxed{T = \frac{2\pi m}{Bq}}$$

or the angular speed (ω) of the particle is, $\omega = \frac{2\pi}{T} = \frac{Bq}{m}$

$\therefore$

$$\boxed{\omega = \frac{Bq}{m}}$$

Frequency of rotation is, $f = \frac{1}{T}$

or

$$\boxed{f = \frac{Bq}{2\pi m}}$$

Following points are worthnoting regarding a circular path :

(i) T, f and ω are independent of v while the radius is directly proportional to v.

(ii) Hence, if two, charged particles of equal mass and charge enter in a magnetic field $\vec{B}$ with different speeds v_1 and v_2 ($> v_1$) at right angles, then

$$T_1 = T_2$$

but $r_2 > r_1$

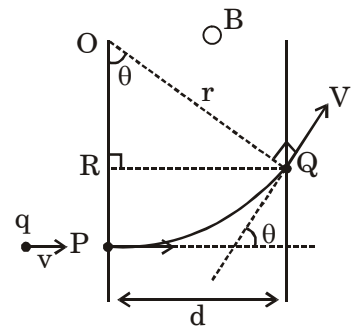
as shown in figure.

Example 1

A charged particle q enters a magnetic field region which spreads over a length d with a velocity v. Find the angular and sideways deflection and discuss various cases.

Solution :

$$\left(r = \frac{mv}{qB}\right)$$



Now $(\sin\theta) = \left(\frac{d}{r}\right) = \left(\frac{qBd}{mv}\right)$

$$\therefore\ \theta = \sin^{-1}\left(\frac{qBd}{mv}\right)$$

Side ways deflection, PR $= r - r\cos\theta$

$= r\,(1 - \cos\theta)$ where $\sin\theta = \left(\frac{qBd}{mv}\right)$

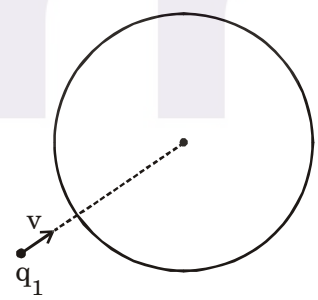
Cases :

(i) When $(d > r)$, then deviation is π.

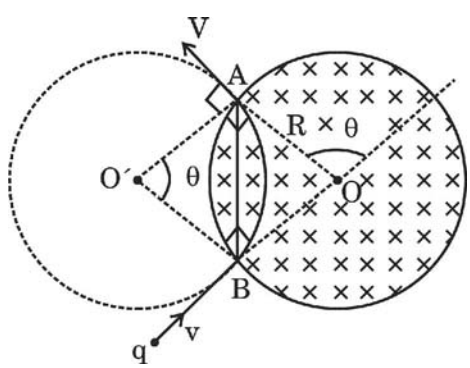
(ii) when $d \simeq r$, then deviation is $\pi/2$.

Example 2

A charged particle is directed towards the centre of a circular field region having magnetic field $\vec{B}$ ◯ and radius R. The charge and the velocity of the particle are q and v respectively. Find the angular deflection of the particle.



Solution :



{Note the path of the given particle}

Here OA = R, radius of given circle.

O´A = r , radius of the curvature of the path made by charged particle.

Where $r = \frac{mv}{qB}$

Now, if we consider ΔOAB and ΔOAB, then length AB in both the triangles will be,

In ΔOAB

$$AB = 2\left(\sin\left(\frac{180-\theta}{2}\right)\right) = 2R\cos\frac{\theta}{2}. \qquad \text{... (1)}$$

In ΔO´AB

$$AB = 2r\sin\left(\frac{\theta}{2}\right). \qquad \text{... (2)}$$

Hence, dividing (1) and (2)

$$\Rightarrow \frac{\theta}{2} = \frac{R}{r} = \frac{Rq\,B}{mv}$$

$$\therefore \boxed{\theta = 2\tan^{-1}\left(\frac{qRB}{mv}\right)}$$

HELICAL MOTION

Case 3 : When $\vec{v}$ makes certain angle with the magnetic field

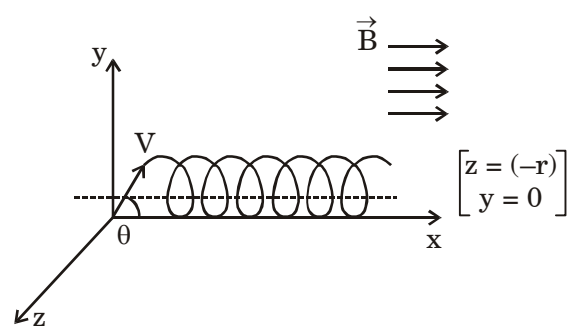
- In this case the motion executed by the charged particle is helical motion.
- In this case, resolve the velocity vector into two components, one parallel to $\vec{B}$ field and second perpendicular to the magnetic field.

$V_{\perp} = v\sin\theta$

$V_{\parallel} = v\cos\theta$

∴ radius of the helix

$$= \frac{mv_{\perp}}{qB} = \left(\frac{mv\sin\theta}{qB}\right)$$



Time period of 1 complete rotation = $\left(\frac{2\pi m}{qB}\right)$

Pitch of the helix = axial distance covered during one complete rotation

$P = V_{||}T$

$$\boxed{P = (V\cos\theta)\left(\frac{2\pi m}{qB}\right)}$$

Few important points to note :

(i) The axis of the helix is always parallel to $\vec{B}$ field.

(ii) The plane of circular motion is perpendicular to the magnetic field lines.

(iii) Pitch of the helix = (v cos θ) T

(iv) A line drawn parallel to axis from the initial point of projection of the particle is tangent to the helix and after making every full rotation, the particle will touch this tangent again and again.

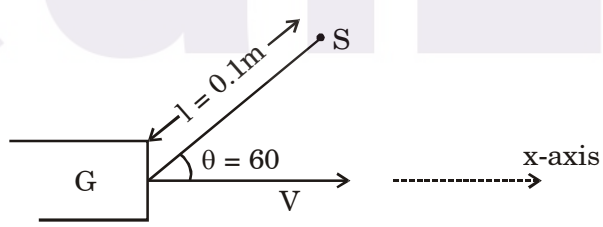
If the tangent is taken along the x-axis, then the particle will hit the axis after every pitch. i.e. it will touch the x-axis at a distance.

$$\boxed{x = np}$$

Example 3

An electron gun emits electrons each with K.E. = 2 KeV and directed along positive x-axis. All electrons are required to hit the spot S, where l = GS = 0.1 m. A uniform magnetic field exists along GS.

Find the minimum valve of B required.



Solution :

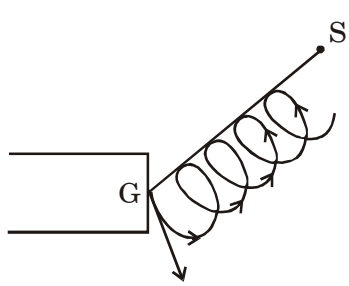
Let the initial velocity v be directed along x-axis,

$\therefore \quad V_{||} = v\cos\theta = v\cos 60$

{This is the velocity parallel to GS}

$V_{\perp} = v\sin\theta = v\sin 60$

{This velocity is perpendicular to GS}



The GS is the tangent line, along with the particle will be touching again and again after every complete rotation

$$n \text{ (pitch of helix)} = GS$$

$$\Rightarrow \quad n\,(v \cos 60 \quad T) = l$$

and
$$v = \sqrt{\frac{2(K.E.)}{m}} = \sqrt{\frac{4000\,e}{m}}\ J,$$

$$T = \frac{2\pi m}{eB}$$

$$\therefore \quad n\left[\frac{v}{2} \times \left(\frac{2m}{eB}\right)\right] = l$$

$$\therefore \quad B = \frac{n\pi m}{el}\sqrt{\frac{4000\,e}{m}}$$

$\therefore$ For minimum value of B, $n = 1$

$$\therefore \quad \boxed{B = \frac{\pi m}{el}\sqrt{\frac{4000\,e}{m}}}$$

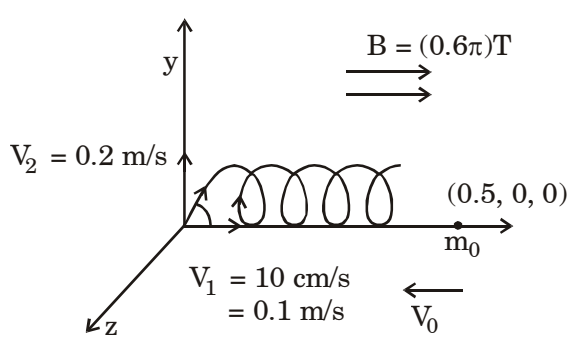
Example 4

A charged particle having charge 10 μC and mass m = 3 mg has velocity 10 (i + 2j) cm/s at origin at t = 0.

Another uncharged particle initially located at (50 cm, 0, 0) is moving with constant velocity along –ve x-axis. The particle collides and stick together and start moving in a circle. Then find the possible values mass of uncharged particle, if a uniform magnetic field, $\vec{B} = 0.6\pi\ T\tilde{i}$ exists in the region.

Also find the radius of combined mass's circular motion.

Solution :



Note that particle 1 follow a helical path with axis of helix parallel to $\vec{B}$ field.

Also, observe that x axis is the tangent line

$\therefore$ When collision takes places the time will be 'n' multiple of time period.

$$\therefore \quad T = \left(\frac{2\pi m}{qB}\right) = \frac{2\pi \times 3 \times 10^{-6}}{10 \times 10^{-6} \times 6\pi \times 10^{-1}} = 1\,s$$

$\therefore$ t_o = time of collision = nT = 'n' seconds

$\Rightarrow$ $(0.5 - V_0\, t_o)$ = n (pitch of helix)

$\Rightarrow$ $(0.5 - V_0 n) = n\,(0.1 \quad T) = 0.1\ n$ .... (1)

After collision, if they move in circle, then conserve momentum along x-axis,

$$m_0 v_0 - m_1 v_1 = 0$$

$$\Rightarrow \boxed{\frac{(3\times 0.1)}{v_0} = m} \quad .... (2)$$

Substitute (2) in (1),

$$(0.5 - 0.1\ n) = V_0\ n = \frac{n\,(3\times 0.1)}{m}$$

$$\therefore \quad m = \frac{0.3\ n}{(0.5-0.1n)} \ (mg)$$

$\therefore$ possible values of n = 1, 2, 3, 4

$\therefore$ possible values of m = $\frac{3}{4}$ mg, 2 mg, $\frac{9}{2}$ mg, 12 mg

Conserve momentum along y-axis, $\left(\frac{m_1 v_2}{m_1 + m_0}\right) = v'$ $\boxed{r = \left(\frac{mv'}{qB}\right)}$

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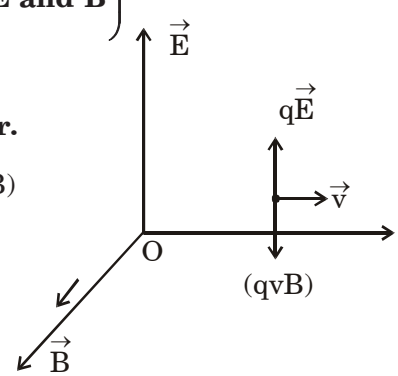
4. Motion of the charged particle in combined field $\left(\vec{E} \text{ and } \vec{B}\right)$

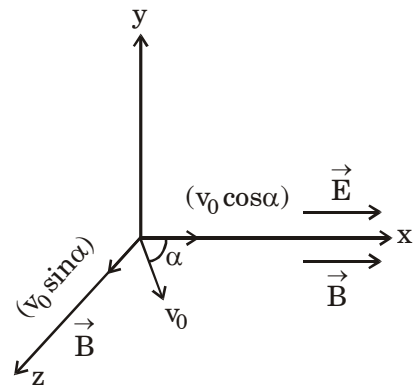


Case I. When $\vec{E}$ and $\vec{B}$ and $\vec{v}$ are mutually perpendicular.

If the particle moves without deflection, then, (qE = qvB)

$$\therefore \quad \boxed{v = \frac{E}{B}}$$

Case II. When $\vec{E}$ and $\vec{B}$ are parallel to each other



$$v_x = (v_0 \cos\alpha) + \left(\frac{qE}{m}\right) t$$

But the $(v_0 \sin\alpha)$ component remains same.

Now, after any time t

$$v_y = (v_0 \sin \alpha) \sin \theta = (v_0 \sin\alpha) \sin (\omega t)$$

$$= (v_0 \sin \alpha) \sin \left(\frac{qBt}{m}\right)$$

$$v_z = (v_0 \sin \alpha) \cos \theta$$

$$= (v_0 \sin \alpha) \cos \omega t = (v_0 \sin \alpha) \cos \left(\frac{qBt}{m}\right)$$

$$\therefore \quad \boxed{\vec{v} = (v_x \tilde{i} + v_y \tilde{j} + v_z \tilde{k})}$$

Radius of curvature of the trajectory $= \dfrac{v^3}{|\vec{a} \times \vec{v}|}$

Where a is the centripetal acceleration.

Note : This is again a type of helical motion but the pitch is not constant.

Case III. Electric and Magnetic field are mutually perpendicular but $\vec{v}$ is not :

This is a general case, when we write the lorentz force equation.

$$\vec{F} = \left[q\vec{E} + q(\vec{v} \times \vec{B}) \right]$$

Assume the velocity at any instant as $\vec{V} = v_x\tilde{i} + v_y\tilde{j} + v_z\tilde{k}$ and then write the indivisual force equation.

$$\therefore \quad \left[m\left(\frac{dv_x}{dt}\right) = F_x \right]; \left[m\frac{dv_y}{dt} = F_y \right]; \left[m\frac{dv_z}{dt} = F_z \right]$$

Example 5

An electron is emitted with negligible speed from the negative plate of a parallel plate capacitor, Charged to potential drop V. Show that the electron will fail to strike the opposite plate if,

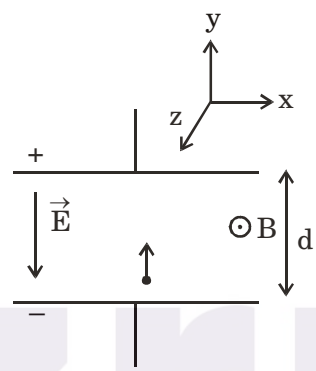
$$d > \left(\frac{2mV}{eB^2}\right)^{\frac{1}{2}} \quad \textbf{[Assume gravity free space]}$$

A magentic field B exists out of plane of paper.

Solution :

Assume at any instant, the velocity acquired by the particle is

$$\vec{v} = v_x\tilde{i} + v_y\tilde{i}$$



($v_z = 0$, $\because$ no force along z-axis)

Write the equation for lorentz force,

$$\vec{F} = \left[q\vec{E} + q(\vec{v} \times \vec{B})\right]$$

$$= \left[(-e)(-E)\tilde{j}\right] + -e\left[\left(v_x\tilde{i} + v_y\tilde{j}\right) \times B\tilde{k}\right]$$

$$= (eE)\tilde{j} + (-e)\left[\left(Bv_y\right)\tilde{i} - Bv_x\tilde{j}\right]$$

$$= -\left(eBv_y\right)\tilde{i} + (eE + eBv_x)\tilde{j}$$

Now, use

$$m\left(\frac{dv_x}{dt}\right) = F_x; \qquad m\left(\frac{dv_y}{dt}\right) = F_y$$

$$\left(\frac{dv_x}{dt}\right) = \left(\frac{-eBv_y}{m}\right) \quad \text{... (1)} \qquad \left(\frac{dv_y}{dt}\right) = \left(\frac{eE + Bv_x}{m}\right) \quad \text{... (2)}$$

Now, there are two methods which can be used :

(i) **work energy method :**

Since, the work done by the magnetic field is zero, the net change in velocity occurs due to electric field only.

$$\boxed{\frac{1}{2}m\left(v_x^2 + v_y^2\right) = (eE)y} \quad \text{... (3)}$$

Now, using eqn. (1),

$$\int_0^{v_x} dv_x = -\left(\frac{eB}{m}\right) . \int v_y \, dt$$

$$v_x = -\frac{eB}{m}(y)$$

Substitute v_x is eqn. (3) to get a relation between v_y and y.

$$\left(\frac{e^2B^2}{m^2}\right) y^2 + v_y^2 = \left(\frac{2eE}{m}\right) y$$

$$\boxed{v_y^2 = \left(\frac{2eE}{m}\right) y - \left(\frac{e^2B^2}{m^2}\right) y^2}$$

Put $v_y = 0$, to obtain the y - co-ordinate when the particle has only v_x component of its velocity < d

$$\Rightarrow \; y\left[\frac{2eE}{m} - \left(\frac{e^2B^2}{m^2}\right) y\right] = 0$$

$$\left(\frac{e^2B^2}{m^2}\right) y = \left(\frac{2eE}{m}\right) \qquad \therefore \qquad y = \left(\frac{2mE}{eB^2}\right)$$

Now, y < d [to not hit the opposite plate]

$$\Rightarrow \; \frac{2mE}{eB^2} < d \qquad \because \qquad \left(E = \frac{V}{d}\right)$$

$$\Rightarrow \; \frac{2mV}{eB^2 d} < d \qquad \Rightarrow \qquad \boxed{\left(\frac{2mV}{eB^2}\right)^{\frac{1}{2}} < d}$$

(ii) Differential Method :

$$\left(\frac{dv_y}{dt}\right) = \frac{e(Bv_x + E)}{m} \quad \text{... (1),} \qquad \left(\frac{dv_x}{dt}\right) = -\left(\frac{eBv_y}{m}\right)$$

Differentiate (1), once again,

$$\left(\frac{d^2v_y}{dt^2}\right) = \left(\frac{eB}{m}\right)\left(\frac{dv_x}{dt}\right)$$

$$\left(\frac{d^2v_y}{dt^2}\right) = -\left(\frac{eB}{m}\right)\left(\frac{eBv_y}{m}\right) = -\left(\frac{e^2B^2}{m^2}\right)v_y$$

Solution of the above Differential equation is,

$$v_y = -A \sin(\omega t + \alpha) \text{ where } \omega = \left(\frac{eB}{m}\right)$$

To obtain the values of α,

Put $t = 0$,

$v_y = -A \sin(\alpha) = 0$ $\quad\quad \therefore (v = 0 \text{ at } t = 0)$

$(\alpha = 0, \pi)$

$$-\left(\frac{dv_y}{dt}\right) = -\omega A \cos(\omega t + \alpha)$$

At $t = 0$, $a = \left(\frac{-eE}{m}\right)$ $\quad\quad [\because v = 0, F_B = 0]$

$$\frac{eE}{m} = +\left(\frac{eB}{m}\right)^A \cos\alpha$$

$$\therefore \quad A = \left(\frac{E}{B}\right) \quad (\alpha = 0)$$

$\Rightarrow$ Hence $\quad v_y = -\frac{E}{B}\sin(\omega t)$

or $\quad \frac{dy}{dt} = \frac{-E}{B}\sin\omega t \quad \Rightarrow \quad y = \frac{+E}{B\omega}\cos\omega t$

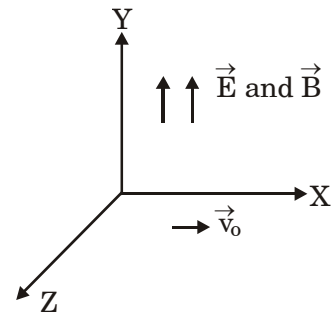
$\Rightarrow \quad y = \frac{vm\cos\omega t}{dB(eB)}$, $\quad$ Now $y < d \quad \Rightarrow \quad \frac{vm}{eB^2} < d^2$

Example 6

A particle of mass m and charge q is moving in a region where uniform, constant electric and magnetic fields $\vec{E}$ and $\vec{B}$ are present. $\vec{E}$ and $\vec{B}$ are parallel to each other. At time $t = 0$, the velocity $\vec{v_0}$ of the particle is perpendicular to $\vec{E}$ (Assume that its speed is always << c, the speed of light in vacuum). Find the velocity $\vec{v}$ of the particle at time t. You must express your answer in terms of t, q, m, the vectors $\vec{v_0}$, $\vec{E}$ and $\vec{B}$ and their magnitudes v_0, E and B.

Solution :

Let us assume, $\tilde{j} = \frac{\vec{E}}{E}$ or $\frac{\vec{B}}{B}$; $\tilde{i} = \frac{\vec{v_o}}{v_o}$ and $\tilde{k} = \frac{\vec{v_o} \times \vec{B}}{v_o B}$



Force due to electric field will be along y-axis. Magnetic force will not affect the motion of charged particle in the direction of electric field (or y-axis). So,

$$a_y = \frac{F_e}{m} = \frac{qE}{m} = \text{constant}$$

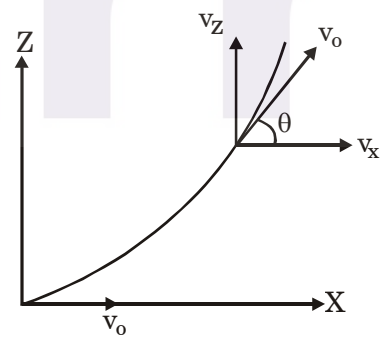
Therefore, $$v_y = a_y\ t = \frac{qE}{m} . t \qquad \text{... (i)}$$

The charged particle under the action of magnetic field describes a circle in x–z plane (perpendicular to $\vec{B}$) with

$$T = \frac{2\pi m}{Bq} \quad \text{or} \quad \omega = \frac{2\pi}{T} = \frac{qB}{m}$$

Initially (t = 0) velocity was along x-axis. Therefore, magnetic force $(\vec{F_m})$ will be along positive z-axis $[\vec{F}_m = q(\vec{v_o} \times \vec{B})]$. Let it makes an angle θ with x-axis at time t, then θ = ωt.

$$\therefore \quad v_x = v_0 \cos\omega t = v_0 \cos\left(\frac{qB}{m}t\right) \qquad \text{... (ii)}$$



and $$v_z = v_0 \sin\omega t = v_0 \sin\left(\frac{qB}{m}t\right) \qquad \text{... (iii)}$$

From Eqs. (i), (ii) and (iii),

$$\therefore \quad \vec{v} = v_x\tilde{i} + v_y\tilde{j} + v_z\tilde{k}$$

$$\therefore \quad v = v_0 \cos\left(\frac{qB}{m}t\right)\left(\frac{\vec{v_o}}{v_o}\right) + \frac{qE}{m}t\left(\frac{\vec{E}}{E}\right) + v_o \sin\left(\frac{qB}{m}t\right)\left(\frac{\vec{v_o} \times \vec{B}}{v_o B}\right)$$

$$\text{or} \quad \vec{v} = \cos\left(\frac{qB}{m}t\right)(\vec{v_o}) + \frac{q}{m}t(\vec{E}) + \sin\left(\frac{qB}{m}t\right)\left(\frac{\vec{v_o} \times \vec{B}}{B}\right)$$

The path of particle will be a helix of increasing pitch. The axis of the helix is parallel to y-axis.

Example 7

A positively charged particle is moving in the x – y plane along the x – axis in the +x direction. After the point P uniform $\vec{E}$ and $\vec{B}$ are present. The path of the particle becomes curved and non-circular. Path remains in the x – y plane. Which one of the following combinations of $\vec{E}$ and $\vec{B}$ is possible.

(a) $\vec{E} = 0;\ \vec{B} = b\tilde{j} + c\tilde{k}$

(b) $\vec{E} = a\tilde{i};\ \vec{B} = c\tilde{k} + a\tilde{i}$

(c) $\vec{E} = 0;\ \vec{B} = c\tilde{j} + b\tilde{k}$

(d) $\vec{E} = a\tilde{i};\ \vec{B} = c\tilde{k} + a\tilde{j}$

Solution :

(a) and (c) are not possible as $\vec{E} = 0$ will create a circular path.

(d) option is also incorrect because in this case force has a z-component.

Separation of positive and negative charges by the help of magnetic field

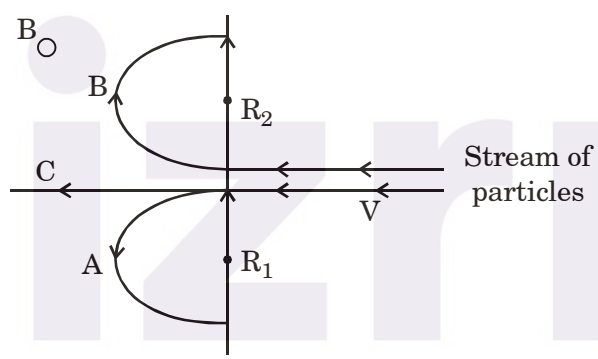
A stream of positive and negative particles moving with a velocity v is incident on the boundary of uniform magnetic field.

$A \equiv (+q_1\ ,\ m_1)$

$B \equiv (-q_2\ ,\ m_2)$

$C \equiv$ neutrons

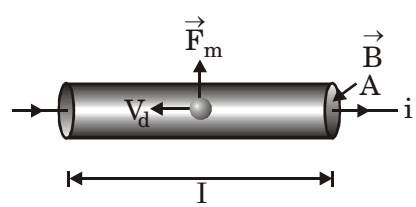
$R_1 = \dfrac{m_1 v}{q_1 B}\ ,\ R_2 = \dfrac{m_2 v}{q_2 B}$



The positive and negative charges will move on circles of different radius as shown in the diagram. The neutral particles neutrons will have zero deviation.

5. Magnetic Force of A Current Carrying Conductor

A charged particle in motion experiences a magnetic force in a magnetic field. Similarly, a current carrying wire also experiences a force when placed in a magnetic field. This follows from the fact that the current is a collection of many charged particles in motion. Hence, the resultant force exerted by the field on the wire is the vector sum of the individual forces exerted on all the charged particles making up the current. The force exerted on the particles is transmitted to the wire when the particles collide with the atoms making up the wire.



Suppose a conducting wire carrying a current i, is placed in a magnetic field $\vec{B}$. The length of the wire is l and area of cross-section is A. The free electrons drift with a speed v_d opposite to the direction of current. The magnetic force exerted on the electron is,

$$d\vec{F}_m = -e(\vec{v}_d \times \vec{B})$$

If n be the number of free electrons per unit volume of the wire, then total number of electrons in volume Al of the wire are, nAl. Therefore, total force on the wire is,

$$\vec{F}_m = -e(nAl)(\vec{v}_d \times \vec{B})$$

If we denote the length l along the direction of the current by $\vec{l}$, then the above equation becomes

$$\boxed{\vec{F}_m = i(\vec{l} \times \vec{B})}$$

where $neAv_d = i$

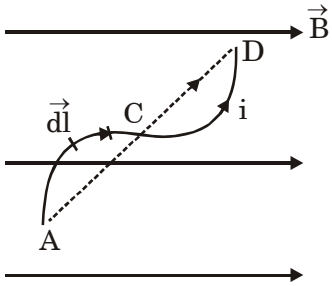
Following points are worthnoting regarding, the above expression :

(i) Magnitude of $\vec{F}_m$ is, $F_m = ilB \sin\theta$, here θ is the angle between $\vec{l}$ and $\vec{B}$. F_m is zero for $\theta = 0$ or 180 and maximum for $\theta = 90$.

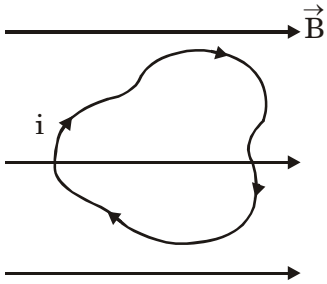
(ii) Here $\vec{l}$ is a vector that points in the direction of the current i and has a magnitude equal to the length.

(iii) The above expression applies only to a straight segment of wire in a uniform magnetic field.

(iv) For the magnetic force on an arbitarily shaped wire segment, let us consider the magnetic force exerted on a small segment of vector length $d\vec{l}$.



(a)



(b)

$$d\vec{F}_m = i(d\vec{l} \times \vec{B}) \quad \text{... (ii)}$$

To calculate the total force $\vec{F}_m$ acting on there wire shown in figure, we integrate over the length of the wire.

$$F_m = i\int_A^D (d\vec{l} \times \vec{B}) \quad \text{... (iii)}$$

Now let us consider two special cases involving Eq. (iii). In both cases, the magnetic field is taken to be constant in magnitude and direction.

Case 1. A curved wire ACD as shown in Fig. (a) carries a current i and is located in a uniform magnetic field $\vec{B}$. Because the field is uniform, we can take $\vec{B}$ outside the integral in Eq. (iii) and we obtain.

$$\vec{F}_m = i\left(\int_A^D d\vec{l}\right) \times \vec{B} \quad \text{... (iv)}$$

But the quantity $\int_A^D d\vec{l}$ represents the vector sum of all length elements from A to D. From the polygon law of vector addition the sum equals the vector directed from A to D. Thus,

$$\vec{F}_m = i(\vec{l} \times \vec{B})$$

or we can write, $\vec{F}_{ACD} = \vec{F}_{AD} = i(\vec{AD} \times \vec{B})$ in uniform field.

Case 2. An arbitrarily shaped closed loop carrying a current i is placed in a uniform magnetic field as shown in Fig. (b). We can again express the force acting on the loop in the form of Eq. (iv), but this time we must take the vector sum of the length elements $d\vec{l}$ over the entire loop,

$$\vec{F}_m = i\left(\oint d\vec{l}\right) \times \vec{B}$$

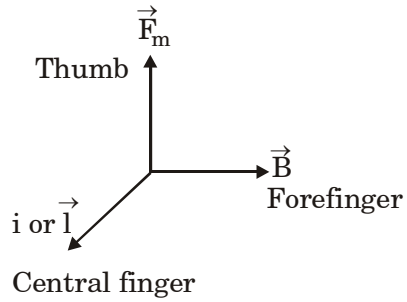
Because the set of length elements forms a closed polygon, the vector sum must be zero.

$$\therefore \quad \vec{F}_m = 0$$

Thus, the net magnetic force acting on any closed current loop in a uniform magnetic field is zero.

(v) The direction of $\vec{F}_m$ can be given by **Fleming's left hand rule**

According to this rule, the forefinger, the central finger and the thumb of the left hand are stretched in such a way that they are mutually perpendicular to each other. If the central finger shows

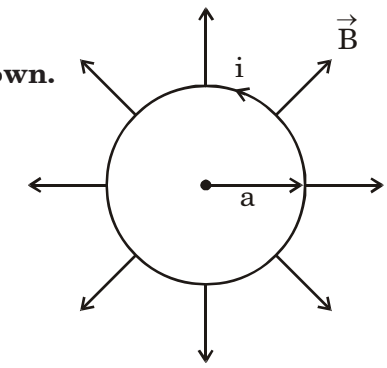


the direction of current (or $\vec{l}$) and forefinger shows the direction of magnetic field ($\vec{B}$), then the thumb will give the direction of magnetic force ($\vec{F}_m$).

(vi) The point of application of magnetic force is taken to be centre of Mass of the body.

Example 8

Find the magnetic force on the ring when $\vec{B}$ is directed as shown.



Solution :

The force on an element will be given by

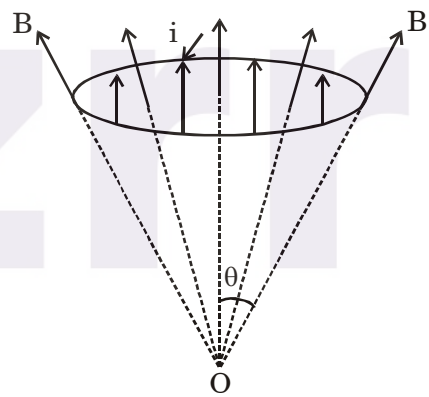
$$d\vec{F} = i(\vec{dl} \times \vec{B})$$

$$= idlB \sin\left(\frac{\pi}{2}\right) \qquad \left\{\begin{array}{l}\text{since, } \vec{B} \text{ is always perpendicular} \\ \text{to any element}\end{array}\right\}$$

$$\therefore F = iB(2\pi a)$$

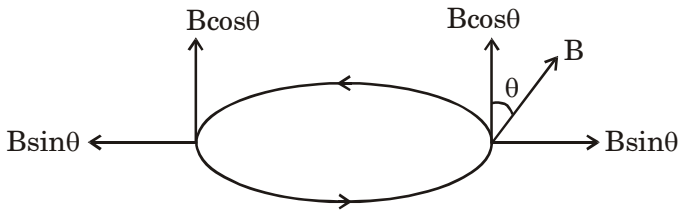
Example 9

The magnetic field is directed as shown. Then find the magnetic force on the ring (Radius = a, current = i)



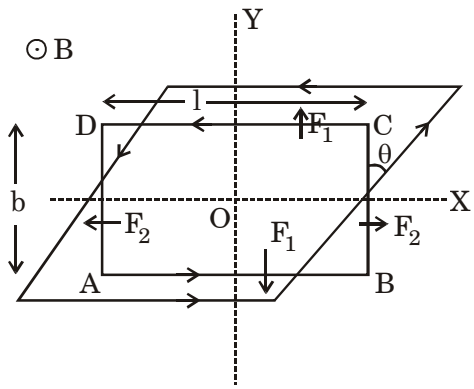
Solution :

Resolve each of field lines, one perpendicular to the plane and the other along the plane of ring. The force due to all of the Bcos θ component will cancel out and the net force will be downward due to B sin θ component



$$F = i(B \sin\theta)\, 2\pi a$$

6. Torque Experienced by a current loop in uniform field



(i) When the plane of the loop is placed perpendicular to the field lines :

$F_1 = iBl\ (\tilde{j})$ $\qquad$ $F_1' = iBl\ (j)$

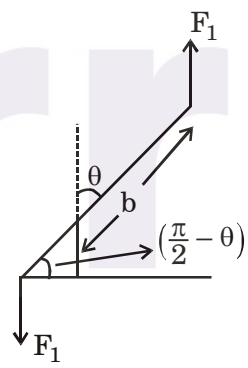
$F_2 = iBb\ (-\tilde{i})$ $\qquad$ $F_2' = iBb\ (\tilde{j})$

Hence, net force = 0 and net couple = 0 (line of action of the forces coincide).

(ii) Now, the plane of the coil is rotated by an angle θ with the vertical

Net force still remain zero

But, now, a couple is formed due to length of wire AB and CD.



$$\vec{\tau} = \vec{F} \times (\text{perpendicular distance})$$

$$= [iBl]\ b \cos\left[\frac{\pi}{2} - \theta\right]$$

$$= iBlb \sin\theta$$

$$\therefore \quad \boxed{\vec{\tau} = i\ BA \sin\theta}$$

Note : In this case, the coil rotates about on axis passing through its CM, ∵ there is no translational motion of the CM of the body.

∴ torque experienced by the loop = $\vec{\tau} = i\,(\vec{A} \times \vec{B})$

where $\vec{A} \equiv$ area of the loop

$\vec{B} \equiv$ magnetic field.

If N turns are present in the loop,

$\vec{\tau} = Ni\,(\vec{A} \times \vec{B})$ or $\boxed{\vec{\tau} = (\vec{M} \times \vec{B})}$

where $\vec{M} \equiv$ magnetic moment of the loop = $(i\vec{A}N)$.

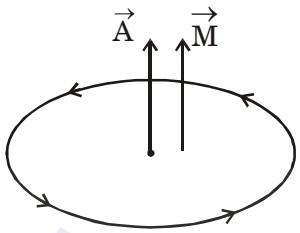
Importance of Number of Turns

If the coil has uniform binding and the size and the shape also remain uniform, then n turns means n identical loops placed one after another very close to each other. {(ni) means effective current}.

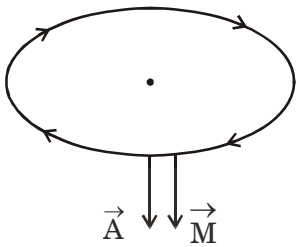
Area Vector

The area vector of a loop is perpendicular to the plane of loop. The magnitude of area vector is equal to the area of loop.

Sign Convention



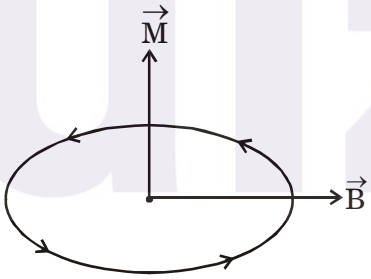
Anti clock wise

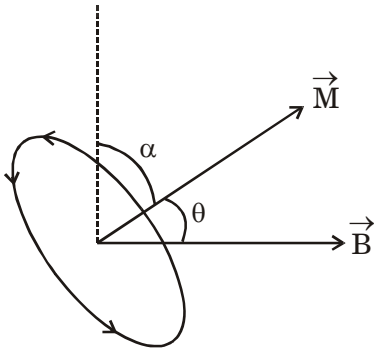


clock wise current

7. Potential Energy

When the magnetic moment is perpendicular to the loop is said to be in the **zero energy state.**





When the loop rotates due to the torque given by magnetic field. Then the rotation K.E. of the loop increases and the P.E. decreases.

Calculating the work done by Magnetic Field :

$$W_B = -\int \tau . \, d\theta$$

$$= -\int_{\frac{\pi}{2}}^{\theta} MB \sin\theta \, d\theta$$

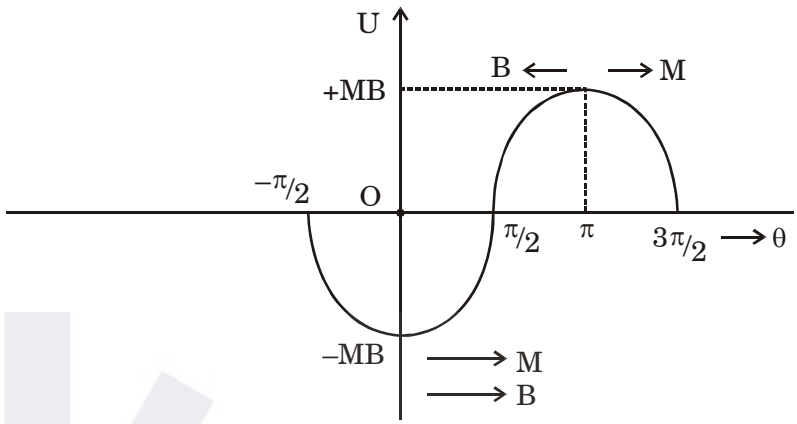
$$W_B = MB \cos \theta$$

Thus, the potential energy will decrease by an amount MB cos θ, Therefore

$$U = -MB\cos\theta$$

or $$\boxed{U = \vec{M}\cdot\vec{B}}$$

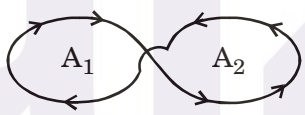
Graph of U v/s θ :



When $\vec{M}$ and $\vec{B}$ are in the same direction then the loop is in stable equilibrium state. And when they are in opposite direction then the loop is in unstable equilibrium state.

Calculating Magnetic Dipole Moment vector in different cases :

Case 1 :

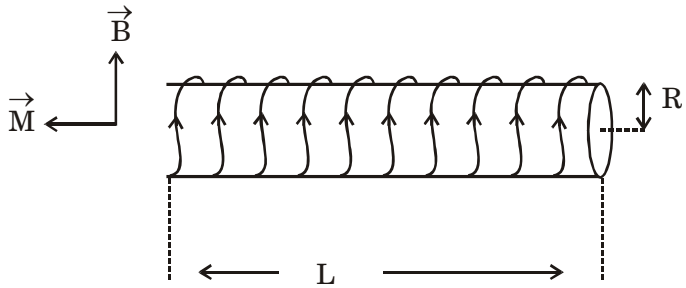


$$M = (A_2 - A_1)$$

In this case if $A_1 = A_2$, then inside a uniform magnetic field the net torque and the net force acting on the loop will be zero. The loop will be in the state of neutral equilibrium.

Case 2 :

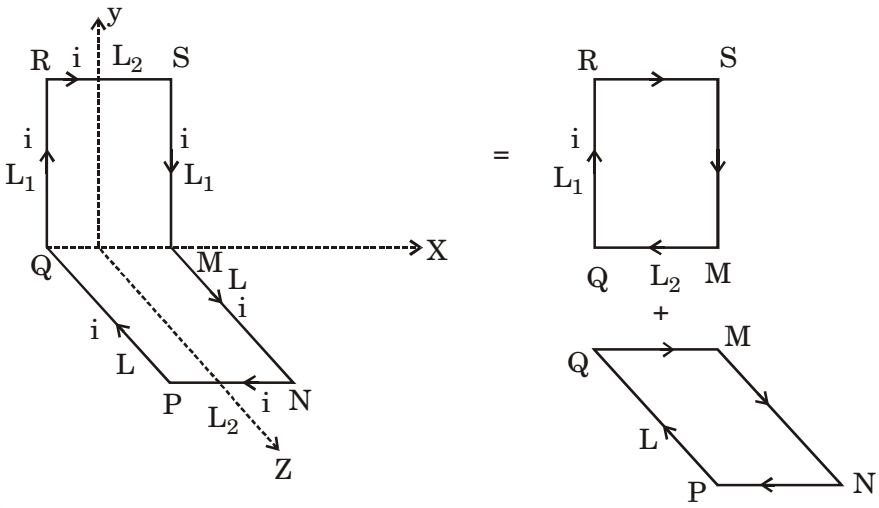
Solonoid : Consider a solonoid of length l and radius R. Let n be the number of turns per unit length



$$[M = (nL)\, i\, \pi R^2]$$

Case 3 :

3 Dimensional Loop



The 3 Dimensional loop can be divided into two 2 dimensional loop. When we join the segment QM of both such loop their currents cancel out and we get the original loop.

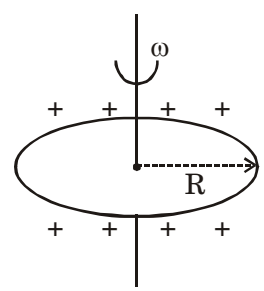
$$\vec{M}_1 = -iL_1 L_2 \tilde{k}$$

$$\vec{M}_2 = -iLL_2 \tilde{j}$$

The net dipole moment vector of the 3 D loop will be $\vec{M}_1 + \vec{M}_2$.

Calculating The Magnetic Dipole Moment of Rotating Charges

In the diagram shown the ring carrys uniformly distrubed charge Q and it is rotating with constant angular velocity ω.



Time taken by the ring to complete one revolution will be $\frac{2\pi}{w}$.

With respect to an observer standing in the state of rest, the charge Q will cross him in $\frac{2\pi}{w}$.

$$i = \frac{\theta}{2\pi / w}$$

Magnetic moment $M = \pi R^2 i = \pi R^2 \frac{\theta}{2\pi / w}$

If the ring has uniformly distributed mass m, then the angular momentum of the ring will be MR^2W.

$$L = Iw$$

$$L = mR^2\omega$$

$$\Rightarrow \boxed{\frac{M}{L} = \frac{Q}{2m}}$$

*If mass and charge are both distributed in the same manner then the value of $\frac{\mu}{L}$ is always $\frac{\theta}{2M}$.

Magnetic Dipole moment of A uniformly charged disc

Consider a disc of radius R carrying uniformly distributed charge Q on it. We have to find out magnetic moment.

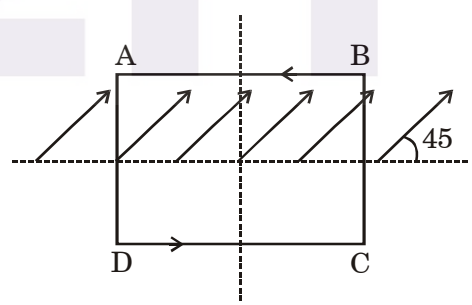
$$\frac{M}{L} = \frac{Q}{2m}$$

$$\frac{M}{\frac{mR^2}{2}w} = \frac{Q}{2m}$$

$$\boxed{M = \frac{QwR^2}{4}}$$

Example 10

A square current loop is placed in a uniform magnetic field as shown which makes an angle of 45 with the x-axis. Mass of each wire side = m, and length l.



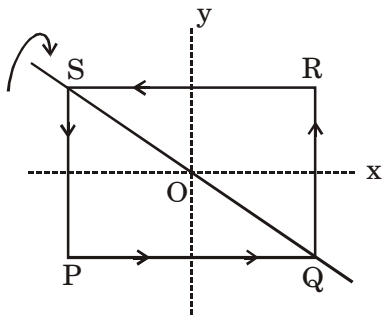
Find :

(i) Net torque acting on the loop and find its direction.

(ii) Angular displacement in the small time t. Neglect the variation in torque during the entire time instant.

Solution :

$$\vec{B} = \frac{B}{\sqrt{2}}\left(\tilde{i} + \tilde{j}\right), \quad \vec{M} = il^2(\tilde{k})$$



$$\therefore \quad \vec{\tau} = (\vec{M} \times \vec{B})$$

$$= \frac{iBl^2}{\sqrt{2}}\left[(\tilde{i} + \tilde{j}) \times \tilde{k}\right]$$

$$\vec{\tau} = \frac{iBl^2}{\sqrt{2}}\left[(\tilde{i} - \tilde{j})\right]$$

$\therefore \quad |\vec{\tau}| = iBl^2$

$\because$ torque is constant,

$$\therefore \quad \alpha = \left(\frac{\tau}{I}\right) = \frac{iBl^2}{I_{QS}} = \left(\frac{3iB}{2m}\right)$$

$$\left[I_{QS} = \frac{1}{2}\left[\left(\frac{ml^2}{12}\right) + \left(\frac{ml^2}{4}\right) \times 4\right] = \frac{2}{3}\, ml^2\right]$$

$\therefore$ Angular displacement in the time t

$$\theta = \frac{1}{2}\alpha\, \Delta t^2 = \frac{1}{2}\left(\frac{3iB}{2m}\right)\Delta t^2 = \left(\frac{3iB}{4m}\right)\Delta t^2$$

Note : In the question if do not neglect the variation in the torque, then,

$$\tau = (MB\cos\theta)$$

$$w\left(\frac{dw}{d\theta}\right) = \frac{MB\cos\theta}{I}$$

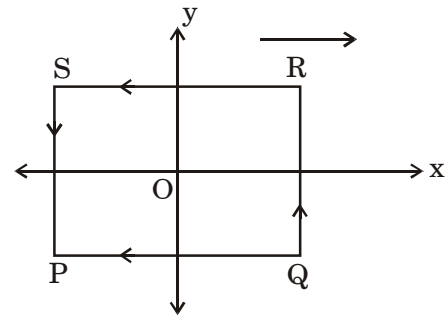
$$\int_0^w (wdw) = \frac{MB}{I}\int_0^\theta \cos\theta\, d\theta$$

$$\therefore \quad \left(\frac{w^2}{2}\right) = \frac{MB}{I}\sin\theta \qquad \therefore \quad \boxed{w = \sqrt{\frac{3iB}{M}\sin\theta}}$$

Example 11

A current carrying loop is placed on a horizontal plane as shown, and each side wire has a length l and total mass of the frame = m. Find the minimum value of current, so that it starts toppling. A horizontal magnetic field B exists the x-axis.

Solution : Magnetic moment is given by :



$$\vec{M} = il^2\tilde{k}$$

$$\vec{B} = B\tilde{i}$$

thus direction of torque is along + y-axis.

$\therefore$ the square loop tries to rotate about the edge RQ.

To prevent the square loop from toppling, the couple must be less than the weight of the torque. If this couple is able to lift the loop from its initial position then, it will certainly topple the frame,

$$\tau_{mg} = mg\,\frac{l}{2}\,\sin\theta \text{ will go on decreasing.}$$

In critical conditions

$$\vec{\tau}_B = \vec{\tau}_{mg}$$

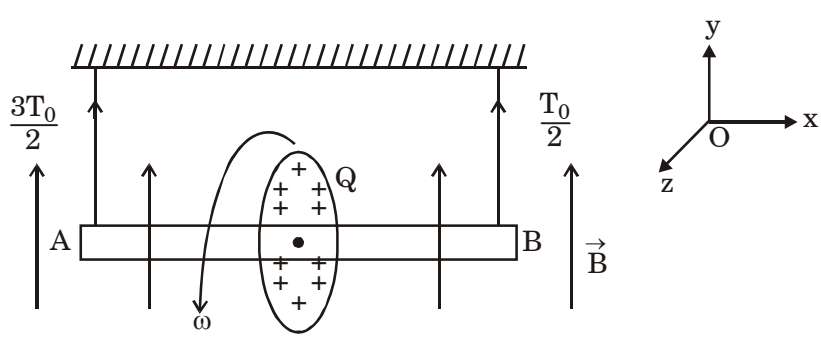
$$\Rightarrow \quad Bil^2 = mg\left(\frac{l}{2}\right)$$

$\therefore$ to just start topping, $\quad i \geq \left(\frac{mg}{2lB}\right)$

Example 12

A light thin horizontal rod is suspended by 2 identical vertical strings. A non-conducting uniform disc of radius R is mounted on the rod as its axle and has a uniform charge distribution of charge Q along its edge circumference. The disc is rotated with constant ω when a vertical magnetic field is switched on in the region. Find B_{max}, so that the string does not break, if the breaking tension is $\left(\frac{3T_o}{2}\right)$. The initial tension in each is T_o.

Solution :



During initial position

weight of the disc = 2 (tension in each string)

$$mg = 2T_0 \qquad \ldots (1)$$

Current due to rotation of the disc

$$i = \left(\frac{Qw}{2\pi}\right) \quad \therefore \vec{M} = i(\pi R^2)\tilde{i}$$

$$\therefore \quad \vec{\tau} = (\vec{M} \times \vec{B}) = i\pi R^2 B\left[\tilde{i} \times \tilde{j}\right] = i\pi R^2 B\tilde{k}$$

Thus, due to above torque developed, it (rod) tries to rotate about the point A. Thus, the tension in the string A increases but that in string B decreases but the total sum $2T_0$ is constant, since the rod is is equilibrium.

During critical conditions, (balancing the net torque) (Take axis about CM)

$$\left(\frac{3T_0}{2}\right)\left(\frac{l}{2}\right) = \left(i\pi R^2 B\right) + \left(\frac{T_0}{2}\right)\left(\frac{l}{2}\right)$$

$$\Rightarrow \quad \left(T_0 \frac{l}{2}\right) = i\pi R^2 B$$

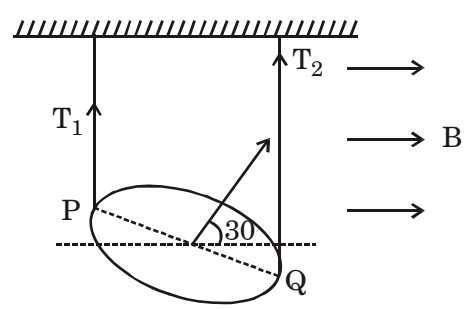
$$\therefore \quad B_{max} = \frac{T_0 l}{2(i\pi R^2)} = \frac{T_0 l(2\pi)}{2Qw\pi R^2}$$

$$\therefore \quad \boxed{B_{max} = \frac{T_0 l}{QwR^2}}$$

Example 13

A circular current loop of radius R carrying current I is suspended by 2 vertical strings with its plane making an angle 60 with the horizontal. Find the tension developed in each string if the loop is in equilibrium.

Solution :



Concept : Resolve the $\vec{M}$ vector into two components.

One along the $\vec{B}$ field [for which $\vec{\tau} = 0$] and other perpendicular to the $\vec{B}$ field.

$\therefore \quad M_{\perp} = (M\sin 30°)$, $M_{||} = M \cos 30$

$$\tau = \vec{M} \times \vec{B} = (M\sin 30°) \times B = \frac{i(\pi R^2)B}{2}$$

The circular loop tries to rotate about the point Q.

$\therefore$ the body is equilibrium, we can balance the forses and the torque acting on the body

$$(T_1 + T_2 = mg) \qquad \text{... (1)}$$

Torque about Q,

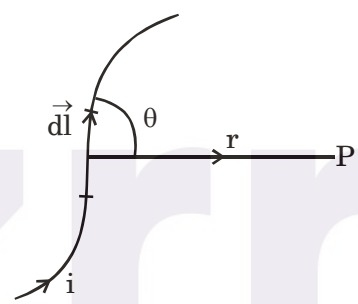
$$[T_1(2R\cos 60°)] + \left[\frac{i\left(\pi R^2\right)B}{2}\right] = mg\ (R \cos 60°) \qquad \text{... (2)}$$

$\Rightarrow$ $$T_1 = mg - i\pi RB$$

and $$T_2 = i\pi RB$$

8. BIOT-SAVART LAW

$$\boxed{\vec{dB} = \frac{\mu_o i}{4\pi} \frac{\vec{dl} \times \vec{r}}{|\vec{r}|^3}}$$



$\vec{r} \equiv$ position vector directed to the point from the current element.

where $\frac{\mu_0}{4\pi} = 10^{-7}\ \frac{T-m}{A}$

Note : Magnitude of $\vec{dB}$ is given by,

$$|\vec{dB}| = \frac{\mu_0}{4\pi} \frac{idl\sin\theta}{r^2}$$

$|\vec{dB}| = 0$ at $\theta = 0$ or 180 and maximum at $\theta = 90$

When the charged particle is moving and the field created (magnetic) due to its motion is given by

$$\vec{dB} = \left(\frac{\mu_0}{4\pi}\right)\left[\frac{q(\vec{v} \times \vec{r})}{r^3}\right] \text{ (q is very small)}$$

$$\therefore \quad \left[\begin{array}{l} i\,\vec{dl} = \left(\dfrac{dQ}{dt}\right)\vec{dl} \\ i\,\vec{dl} = dQ\left(\dfrac{\vec{dl}}{dt}\right) = dQ\,\vec{v} = q\,\vec{v} \end{array}\right]$$

Here, μ_0 = permeability of free space

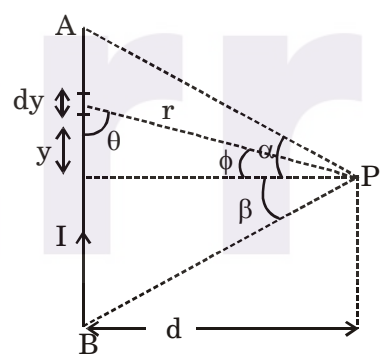
$$= 4\pi \times 10^{-7}\ \frac{T-m}{A}$$

Application of Biot-Savart Law

(A) Straight conductor.

According to Biot-Savart Law,

$$\vec{B} = \frac{\mu_0}{4\pi}\int i\ \frac{\vec{dl}\times\vec{r}}{r^3} \qquad \text{... (i)}$$



Here, every element of the wire contributes to $\vec{B}$ in the same direction (which is into the plane of paper)

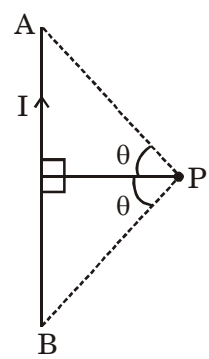
thus, $B = \dfrac{\mu_0}{4\pi}\displaystyle\int\frac{idl\sin\theta}{r^2} = \frac{\mu_0 i}{4\pi}\int\frac{dy\sin\theta}{r^2}$

$y = d\tan\phi$ or $dy = d\sec^2\phi d\phi$

$r = d\sec\phi$ and $\theta = 90 - \phi$

$$B = \frac{\mu_0 i}{4\pi d}\int_{-\beta}^{\alpha}\cos\phi\ d\phi \quad \Rightarrow \quad \boxed{\mathbf{B = \frac{\mu_0 i}{4\pi d}\left(\sin\alpha + \sin\beta\right)}}$$

Various Cases : [Similar to that discussed in case of electro statics]



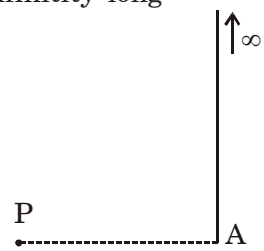
(1) point lying on the perpendicular bisector of the wire.

$\theta_1 = \theta_2 = \theta$

$$B = \frac{\mu_0 i}{4\pi d}\,|\sin\theta|$$

$$B = \frac{\mu_0 i}{2\pi d}\sin\theta$$

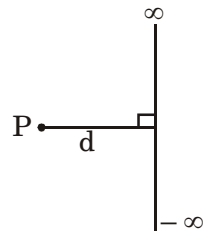
(2) Point lying on the perpendicular of one end when the other end is infinitity long



$$\theta_1 = 0, \quad \theta_2 = \frac{\pi}{2}$$

$$B = \frac{\mu_0 i}{4\pi d} \left|\sin\theta\right|_0^{\pi/2} = \left(\frac{\mu_0 i}{4\pi d}\right)$$

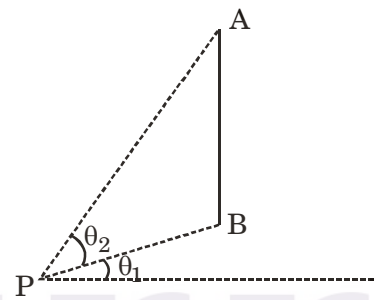
(3) Magnetic field at a distance d from a infinitely long wire



$$\theta_1 = \theta_2 = \frac{\pi}{2}$$

$$B = \frac{\mu_0 i}{4\pi d} \left|\sin\theta\right|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \left(\frac{\mu_0 i}{2\pi d}\right)$$

(4) Magnetic field at a distance d when the given point lies on the same side of the wire.

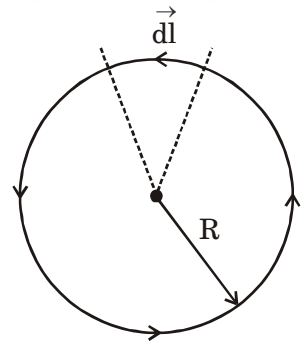


$$B = \frac{\mu_0 i}{4\pi d} \left|\sin\theta\right|_{\theta_1}^{\theta_2} = \left(\frac{\mu_0 i}{2\pi d}\right) \left[\sin\theta_1 - \sin\theta_2\right]$$

Note : Students please take care while taking the angle θ_1 and θ_2. It is the angle made by PB with axis per pendicular to AB

(B) Magnetic Induction at the Centre due to Circular loop

Consider an element $\vec{dl}$ of the ring.



$$\vec{dB} = \frac{\mu_0}{4\pi} \left(\frac{i\,\vec{dl} \times \vec{r}}{r^3}\right)$$

$$\vec{dB} = \frac{\mu_0 i}{4\pi} \left(\frac{dl.\,R}{R^3}\right)$$

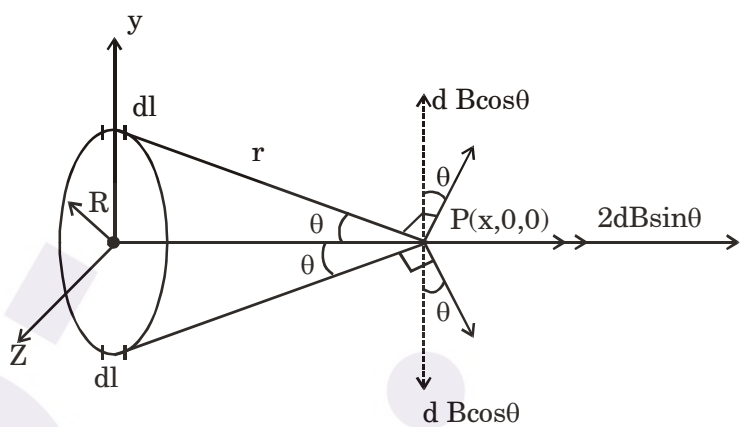
$$B = \frac{\mu_0 i}{4\pi R^2} \int dl = \frac{\mu_0 i}{4\pi R^2} (2\pi R) = \frac{\mu_0 i}{2R}$$

Note : The direction of the $\vec{B}$ field is perpendicular to the plane of the circular loop and given by the right hand rule.

$\therefore$ Magnetic field at the centre of a circular ring = $\left(\frac{\mu_o i}{2R}\right)$

Magnetic field at the centre of a circular arc subtending angle θ at the centre = $\frac{\mu_o i}{2R}\left(\frac{\theta}{2\pi}\right)$

(ii) Magnetic Induction due to a circular loop at an axial point.



We can see from the figure that the vertical component of magnetic fields (due to diameterically opposite lengths dl) will cancel out.

And the net magnetic field will be along x-axis. i.e.

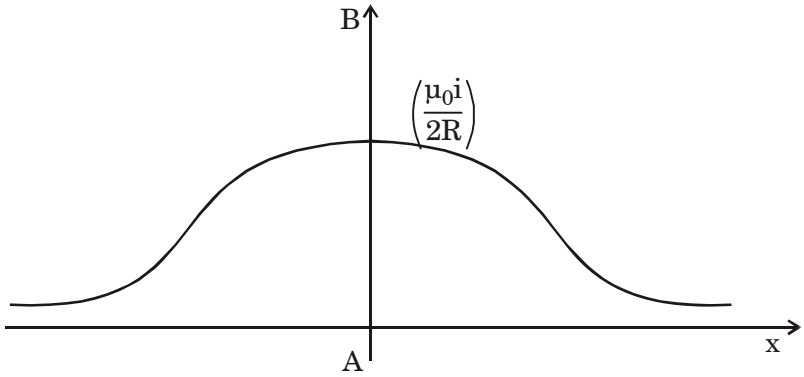
$$B = 2\int dB\sin\theta$$

$$= \int_0^{\pi R} \frac{2\mu_0}{4\pi}\left(\frac{idl \times r}{r^3}\right)\sin\theta$$

$$= \frac{2\mu_0}{4\pi}\frac{R}{r^3}\int_0^{\pi R} dl$$

$$\therefore \quad B = \frac{\mu_0 R^2 i}{2(x^2 + R^2)^{3/2}}$$

Note : The direction of the $\vec{B}$ field at the axial position is same throughout and given by the right hand rule.



In the above case

(i) **When** x **>> R.**

$$B = \frac{\mu_0 i R^2}{2x^3}$$

$$= \frac{\mu_o i R^2}{4\pi x^3}(2\pi) = \left(\frac{\mu_o i}{4\pi}\right)2\left(\frac{i\pi R^2}{x^3}\right)$$

$$B = \left(\frac{\mu_0}{4\pi}\right)\left(\frac{2M}{x^3}\right)$$

[The above result is similar to that of dipoles]

Hence, the above arrangement of circular loop behaves as a magnetic dipole and all the results of the electron dipole are similarly valid.

Note : The above results are valid not only for circular loop but are also valid for any shape of loop when x **>> R.**

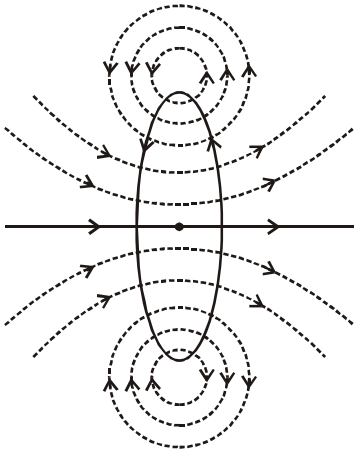
When the dimension of the loop is much smaller in comparison to the distance of the point, then M = iA

$$B = \left(\frac{\mu_0}{4\pi}\right)\left(\frac{2M}{x^3}\right) \text{ at an axial point}$$

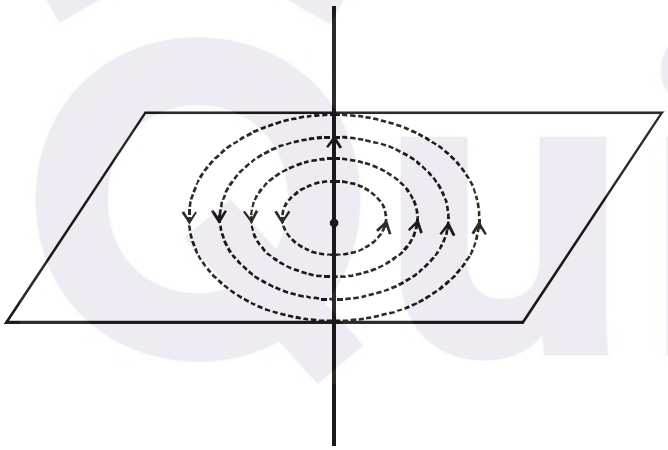
for any point P (r, θ) is space $B = \frac{\mu_o M}{4\pi}\sqrt{3\cos^2\theta + 1}$

PATTERN OF THE FIELD IN VARIOUS CASES

(1) Magnetic field line pattern due to a circular loop.

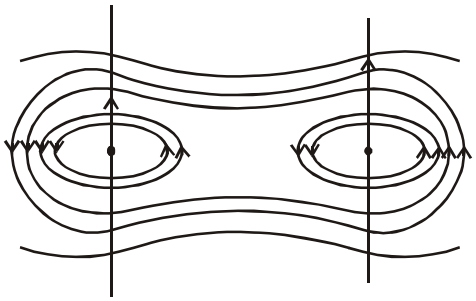


(2) Magnetic field lines are concentric circles centred at the wire and lying in the plane perpendicular to the wire

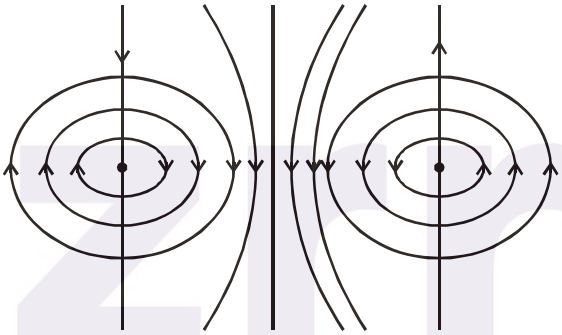


Due to two parallel current wires

(i) **When carrying current in same direction**



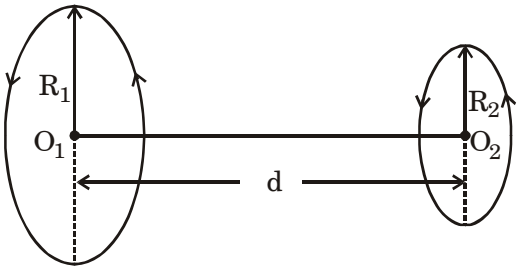
(ii) **When carrying current in opposite direction**



Example 14

Find the force between the two circular loops in given arrangement, if :

(i) $d >> R_2$ **(ii) $d >> R_1, R_2$ for both**



Solution :

(1) **Concept :** For whichever circular loop, d >> R, replace that loop with a dipole. So, in this case replace loop 2 with a magnetic dipole.

$$F = M\left(\frac{dB}{dr}\right), \qquad \text{where} \quad M = i\left(\pi R_2^2\right)$$

Now, $$B = \frac{\mu_0 i R_1^2}{2(R_1^2 + x^2)^{3/2}}$$

where B is the magnetic field the to loop 1 at O_2.

Now, $$\frac{dB}{dx} = \frac{\mu_0 i R_1^2}{2}\left[\frac{\frac{-3}{2}\left(R_1^2 + x^2\right)^{1/2} 2x}{\left(R_1^2 + x^2\right)^3}\right] = -\frac{3}{2}\frac{\mu_0 i R_1^2 x}{\left(R_1^2 + x^2\right)^{5/2}}$$

$$\therefore \quad F = -i\left(\pi R_2^2\right)\left[\frac{3}{2}\frac{\mu_0 i R_1^2 d}{\left(R_1^2 + d^2\right)^{3/2}}\right]$$

The negative sign shows that force is attractive in nature.

(ii) When $d >> R_1, R_2$

In this case both rings are replaced by a dipole.

Now, field due to diple 1 at distance d

$$B_1 = \frac{\mu_0}{4\pi}\left(\frac{2M_1}{x^3}\right)$$

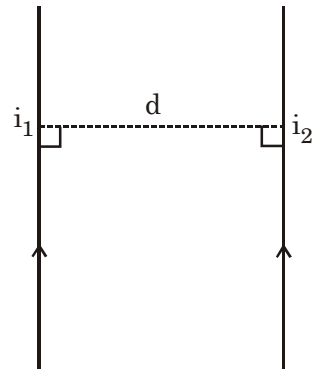
$$\frac{dB_1}{dx} = \frac{\mu_0}{4\pi}\left(-\frac{6M_1}{x^4}\right)$$

$$\therefore \quad F = M_2\left(\frac{dB_1}{dx}\right) = -\frac{\mu_0}{4\pi}\left(\frac{6M_1M_2}{d^4}\right)$$

Force between two parallel current carrying wires.

$$dF = i_2\ dl\left(\frac{\mu_0 i_1}{2\pi d}\right)$$

$$\frac{dF}{dl} = \left(\frac{\mu_0 i_1 i_2}{2\pi d}\right)$$



Thus the force acting per unit length

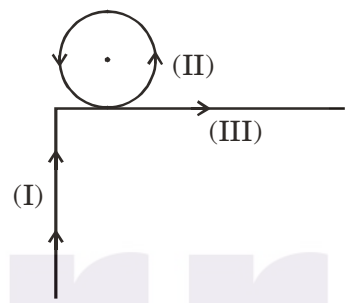
$$F = \left(\frac{\mu_0 i_1\ i_2}{2\pi d}\right)$$

– The force is attraction between the wires carrying current in same direction.

– The force is repulsive between the wire having the current in the opposite direction.

Example 15

A current carrying wire is bent in the form as shown. Field the magnetic field at the centre O of the ring.



Solution :

Let the wires be (I), (II) and ((III) as shown in figure.

Now, Magnetic field due to each of them will be

$$\vec{B}_1 = \frac{\mu_0 I}{4\pi d}\left[1-\sin\frac{\pi}{4}\right]\otimes$$

$$= \frac{\mu_0 I}{4\pi d}\left[1-\frac{1}{\sqrt{2}}\right]\otimes$$

$$\vec{B}_2 = \left(\frac{\mu_0 I}{2R}\right)\otimes$$

$$\vec{B}_3 = \frac{\mu_0 I}{4\pi d}\left[1+\sin\frac{\pi}{4}\right]\odot$$

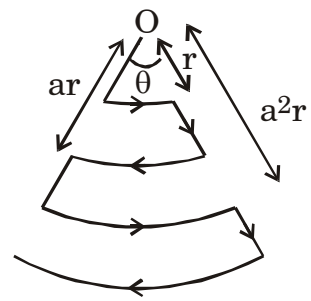
$\therefore$ Net $\vec{B} = \vec{B}_1 + \vec{B}_2 + \vec{B}_3$

$$= \frac{\mu_0 I}{4\pi d}\left[1-\frac{1}{\sqrt{2}}\right] + \frac{\mu_0 I}{2R} - \frac{\mu_0 I}{4\pi d}\left[1+\frac{1}{\sqrt{2}}\right]$$

$$= \frac{\mu_0 I}{4\pi d}\left[\sqrt{2}\right] + \left(\frac{\mu_0 I}{2R}\right)\otimes$$

Example 16

An infinitely long arrangement of wire is laid out in such a way that the radius of each of the circular arc increases 'a' times the previous. Find the $\vec{B}$ at 0.



Solution :

Note that the magnetic field due to straight segment is zero.

$$B_1 = \left(\frac{\mu_0 i}{2r}\right) \cdot \left(\frac{\theta}{2\pi}\right) \odot, \; B_2 = \frac{\mu_0 i}{2(ar)} \frac{\theta}{2\pi} \otimes, \; B_3 = \frac{\mu_0 i}{2(a^2 r)} \cdot \left(\frac{\theta}{2\pi}\right) \odot$$

$\therefore$ Net $\vec{B}$

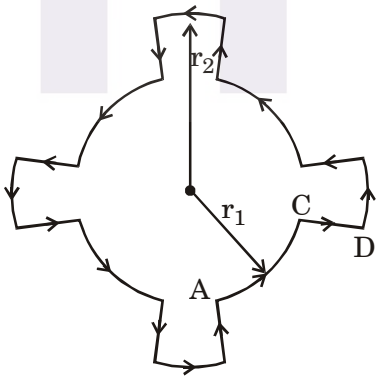
$$= \left[\left(\frac{\mu_0 i}{2r}\right) - \frac{\mu_0 i}{2(ar)} + \frac{\mu_0 i}{2a^2 r} - \frac{\mu_0 i}{2a^3 r} + \ldots \infty\right] \frac{\theta}{2\pi}$$

$$= \frac{\mu_0 i}{2r}\left[1 - \frac{1}{a} + \frac{1}{a^2} - \frac{1}{a^3} + \ldots \infty\right] \frac{\theta}{2\pi}$$

$$= \; = \frac{\mu_0}{2r} \frac{a}{(a+1)} \left(\frac{\theta}{2\pi}\right) \odot$$

Example 17

A circuit consists of 8 alternating arc of radius r_1 and r_2 as shown. Find the $\vec{B}$ field at the centre of the circuit. Also find the force experienced by the arc AC and straight wire CD of the circuit due to current in the straight conductor. Each arc subtends the same angle θ at the centre of the circuit.



Solution :

Since each arc subtends the same angle θ at the centre, hence

$$\theta = \frac{2\pi}{8} = \frac{\pi}{4}$$

Now, magnetic field due to arc's of radius r_1

$$B_1 = \left(\frac{\mu_0 i}{2r_1}\right)\left(\frac{\pi}{4}\right) \times \frac{4}{2\pi} = \frac{\mu_0 i}{4r_1}$$

and magnetic field due to arc's of radius r_2

$$B_2 = \left(\frac{\mu_0 i}{2r_2}\right)\left(\frac{\pi}{4}\right) \times \frac{4}{2\pi} = \frac{\mu_0 i}{4r_2}$$

$\therefore$ Net magnetic field at the centre

$$\vec{B}_{net} = \vec{B}_1 + \vec{B}_2$$

$$= \frac{\mu_0 i}{4}\left\{\frac{1}{r_1} + \frac{1}{r_2}\right\} \otimes$$

(b) The force experienced by the arc AC = 0 {Reason : the magnetic field is in same direction as that of current element}.

The force experienced by the wire CD

$$F = \int_{r_1}^{r_2} \frac{\mu_0 i^2}{2\pi x}\, dx = \frac{\mu_0 i^2}{2\pi} \ln\left(\frac{r_2}{r_1}\right)$$

Example 18

A current (circular) loop is bent along a diameter and placed mutually perpendicular. A charged particle is projected with initial velocity $v_0\left(-\tilde{i}\right)$ and has a charge q and mass m.

(a) Find the instantaneous force experienced by the charge placed at the origin.

(b) If an external field $B\tilde{j}$ is switched on in the region then find the net force experienced by current loop.

Solution :

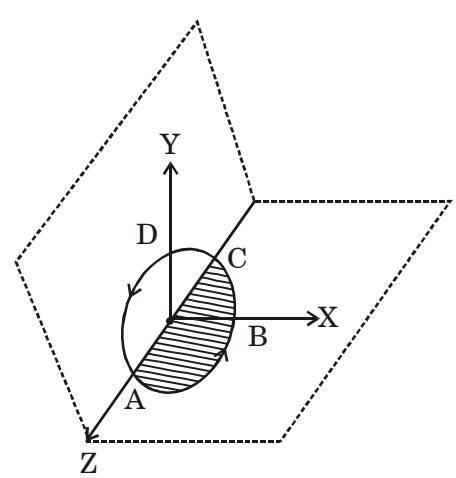
Field due to the circular arc ABC.

$$\vec{B}_1 = \frac{\mu_0 i}{2R}\left(\frac{1}{2}\right)\tilde{j}$$

field due to the circular arc CDA

$$\vec{B}_2 = \frac{\mu_0 i}{4R}\left(\tilde{i}\right)$$

$\therefore$ Net $\vec{B} = \frac{\mu_0 i}{4R}\left(\tilde{i} + \tilde{j}\right)$



$$\therefore \quad \vec{F} = q(\vec{v} \times \vec{B})$$

$$= q\left[v_0(-\tilde{i}) \times \frac{\mu_0 i}{4R}(\tilde{i} + \tilde{j})\right]$$

$$= q\frac{\mu_0 i}{4R} v_0 [-\tilde{k}]$$

(b) force on the arc ABC

$$\vec{l}_{eff} = 2R(-\tilde{k}) \quad \vec{B} = B_0\tilde{j}$$

$$\therefore \quad \vec{F}_1 = I(\vec{l} \times \vec{B})$$

$$= I\left[2R(-\tilde{k}) \times B(\tilde{j})\right]$$

$$= 2\,BRI\,\tilde{i}$$

force on the arc CDA

$$\vec{l}_{eff} = 2R(\tilde{k})$$

$$\vec{F}_2 = I\left[2R\tilde{k} \times B_0\tilde{j}\right]$$

$$= I\,B_0\,2R(-\tilde{i})$$

9. Ampere's Circuital Law

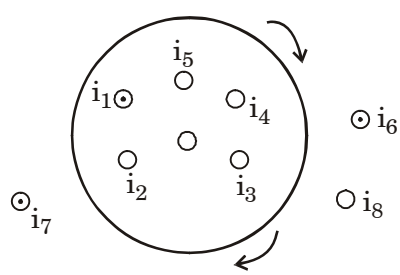
Statement. The circulation of magnetic field along a closed loop is μ_0 times the total current passing through normally, of the area of the closed loop.

$$\boxed{\oint \vec{B}.\vec{dl} = \mu_0 I_{enclosed}}$$

This law is analogous to Gauss Law already covered in electrostatics.

For Example

The figure shows the current carrying conductors enclosed by a closed loop.



Then, $\boxed{\oint \vec{B}.dl = \mu_0[-i_1 + i_2 + i_3 + i_4 + i_5]}$

Important points to note :

(a) The direction of circulation is assumed to be positive and accordingly any current in the direction of circulation will be positive and the other negative.

(b) If the current is at an angle θ with the plane of the loop, then take its two components, one parallel to the loop, and the other perpendicular to the loop. Consider only the normal component in the formula.

(c) The above form of Ampere's Law is not valid when the electric field through the area of loop is not constant.

In that case we need to apply the modified form of Ampere's Law, called **Maxwell Ampere's Law.**

$$\boxed{\oint \vec{B}.\vec{dl} = \mu_0\left[i + \frac{d\phi_E}{dt}\epsilon_0\right]}$$

The term $\frac{d\phi_E}{dt}\epsilon_0$ is called as **displacement current.**

(d) Note that in the Ampere's formula, $\vec{B}$ is due to all the elements in the space, but I contains the summation of the enclosed currents.

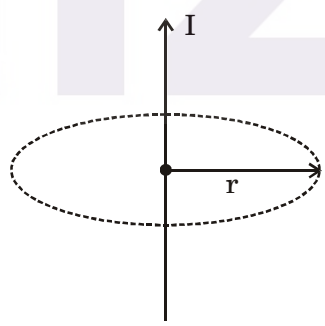
Application of Ampere's Law in calculating the field

(1) **Field due to a infinitely long current carrying wire :**

$$\oint \vec{B}.\vec{dl} = \mu_0 I$$

$$\Rightarrow \quad B\,(2\pi r) = \mu_0 I$$

$$\therefore \quad B = \frac{\mu_0 I}{2\pi r}$$



Note that for a symmetrical field the direction of the $\vec{E}$ field is tangent to the path, whereas it is radial in the case of $\vec{E}$ field,

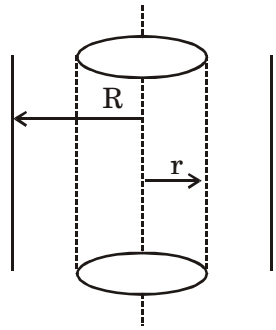
(2) **Field due to along cylindrical conductor of radius R.**

(a) **Field inside the conductor,** $r \leq R$: Consider a closed circular loop at a distance r from central axis.

Now, current enclosed by the loop is $\left(\frac{I}{\pi R^2}\right) \times \pi r^2$

$\therefore$ According to Ampere's Law

$$\oint \vec{B} \cdot \vec{dl} = \mu_0 i$$



$$\Rightarrow \quad B.\, 2\pi r = \mu_0 \frac{Ir^2}{R^2}$$

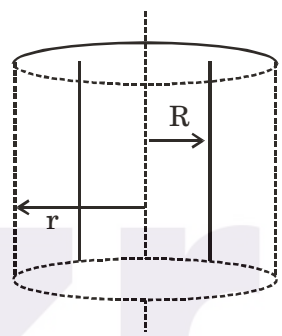
$$\Rightarrow \quad \boxed{B = \frac{\mu_0}{2\pi} \frac{Ir}{R^2}}$$

It can also be written as $\boxed{\vec{B} = \frac{\mu_0}{2} (\vec{j} \times \vec{r})}$

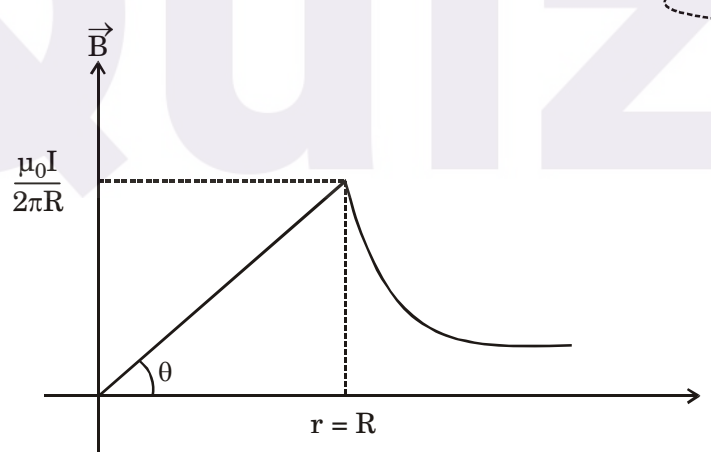
(b) **Outside the cylindrical conductor** (r > R)

Current enclosed by loop = I

$$\oint \vec{B} \cdot \vec{dl} = \mu_0 I$$



$$\Rightarrow \quad \boxed{B = \frac{\mu_0}{2\pi} \frac{I}{r}}$$



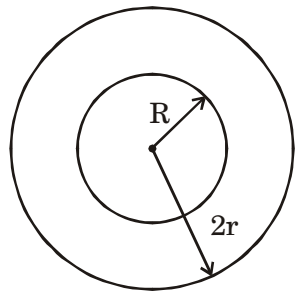
The field outside the cylinder is also given by,

$$\vec{B} = \frac{\mu_0 R^2}{2} = \frac{(\vec{j} \times \vec{r})}{\vec{r}}$$

Where j is the charge density i.e. $\boxed{j = \frac{I}{\pi R^2}}$

Example 19

Find the magnetic field due to a cylindrical shell of inner and outer radii R and 2R with current density $j = \left(\frac{j_0 r}{2R}\right)$ for



(i) $r \leq R$

(ii) $R \leq r \leq 2R$

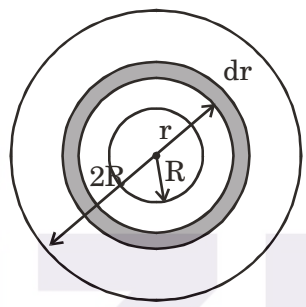
(iii) $2R \leq r$

Here j exists for $R \leq r \leq 2R$

Solution :

(i) for $r \leq R$, $\vec{B} = 0$

(ii) for $R \leq r < 2R$



$$B(2\pi r) = \mu_0 \int_R^r j(2\pi r)\, dr$$

$$B(2\pi r) = \mu_0 \int_R^r \frac{j_0 r}{2R}(2\pi r)\, dr$$

$$B = \frac{\mu_0}{2Rr}\left(\frac{r^3 - R^3}{3}\right) j_0 = \left(\frac{\mu_0}{6R}\right)\left[\frac{r^3 - R^3}{r}\right] j_0$$

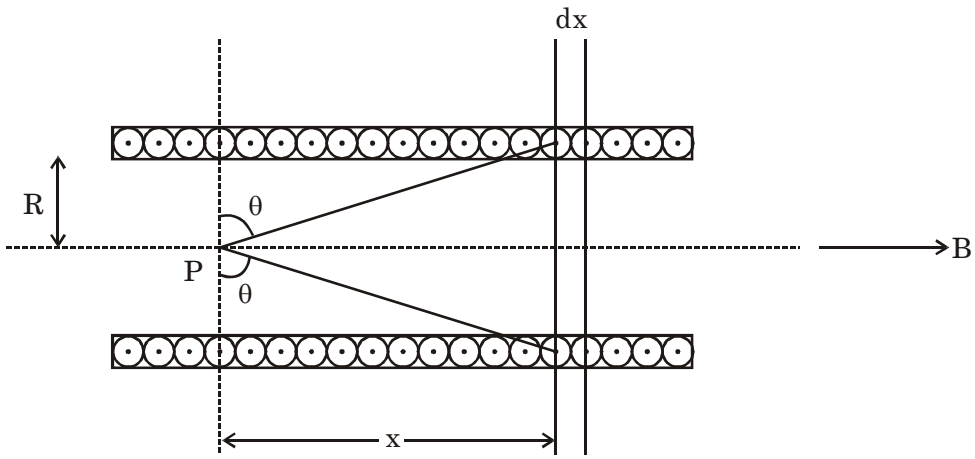
(iii) $r \geq 2R$

$$B(2\pi r) = \mu_0 \int_R^{2R} j(2\pi r)\, dr$$

$$B(2\pi r) = \frac{\mu_0 \pi}{R}\left(\frac{7R^3}{3}\right) j_0$$

$$\therefore \quad \boxed{B = \frac{7}{6}\frac{\mu_0 R^2 j_0}{r}}$$

(3) Solenoid (At any point P on the axial position)



n = number of turns per unit length.

$$dI = i\ (ndx)$$

$$dB = \frac{\mu_0\ dIR^2}{2(x^2 + R^2)^{3/2}}$$

$$dB = \frac{\mu_0\, i(ndx)R^2}{2(x^2 + R^2)^{3/2}}$$

$$\int dB = \int \mu_0 \frac{i(nR^2)\, R \sec^2\theta\, d\theta}{2R^3 \sec^3\theta}$$

$$x = R \tan\theta$$

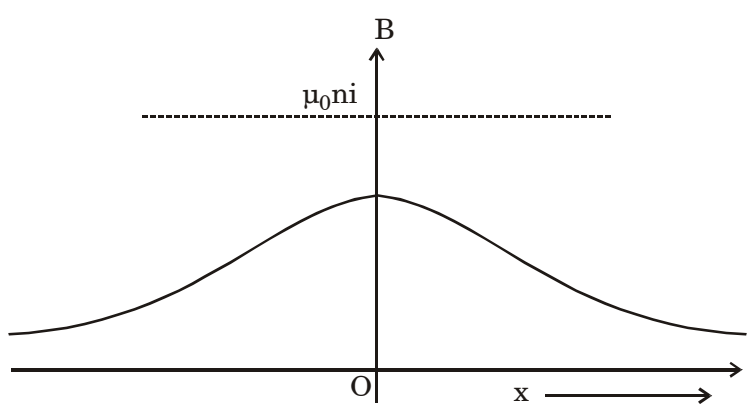
$$dx = R \sec^2\theta\, d\theta$$

$$B = \frac{\mu_0\, in}{2} \int_{-\theta_2}^{\theta_1} \cos\theta\, d\theta$$

$$B = \frac{\mu_0\, in}{2} \left|\sin\theta\right|_{-\theta_2}^{\theta_1}$$

$$\boxed{\therefore B = \frac{\mu_0 in}{2}\left(\sin\theta_1 + \sin\theta_2\right)}$$

At any axial point of the solenoid.



Plot of the B field for O solenoid at an axial position.

For a long solenoid

$\theta_1 = \theta_2 = \pi/2$

$\therefore\ B = \dfrac{\mu_0 \text{ in}}{2}[2] = \mu_0 \text{ in}$

$\therefore \boxed{B = \mu_0 \text{ in}}$

This is the magnitude of $\vec{B}$ field not only at the axial position, but for all positions inside the solenoid.

Note : Outside the solenoid, the magnitude of the $\vec{B}$ field is zero, for a closely wound solenoid.

Reason : The $\vec{B}$ field from the adjacent coilings cancel each other. Hence the net $\vec{B}$ field outside the solenoid is zero.

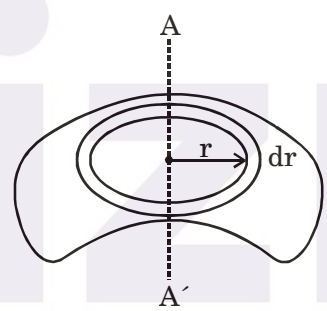
To find magnetic moment of charged bodies rotating about an axis.

$$\mathbf{M} = \int di\mathbf{A}$$

$$di = \frac{Q}{\pi R^2}(rdr)w$$

$$M = \int_0^R \frac{Qw}{\pi R^2}(rdr)(\pi r^2)$$

$$= \frac{Qw}{R^2}\left(\frac{R^4}{4}\right) = \frac{QwR^2}{4}$$



Alternative Shortkut Method

$$\boxed{\textbf{Magnetic Moment} = \frac{q}{2m}.\ \{\text{Mechanical moment (angular momentum)}\}}$$

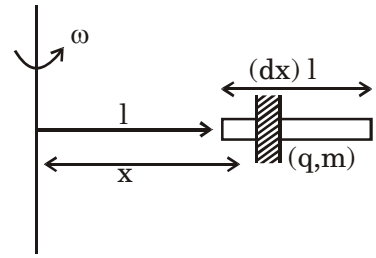
Note : This formula is true always, provided the charge and the mass distribution are identical about the axis.

Example 20

A uniformly charged non-conducting rod is placed in a horizontal plane at a distance l **from the origin and rotated about a vertical axis on the plane passing thorough the origin. Find the magnetic induction at the origin and the magnetic moment of the rod.**

Solution :

Consider an element dx at the distance x.



$$\lambda = \left(\frac{q}{l}\right) = \text{charge per unit length}$$

$$dq = (\lambda dx)$$

Note: The small element (dx) carrying the charge dq behaves a circular loop current.

$$dB = \left(\frac{\mu_0 di}{2x}\right) = \frac{\mu_0\, dq}{(2x)}\left(\frac{\omega}{2\pi}\right)$$

$$B = \frac{\mu_0 w\lambda}{4\pi}\int_l^{2l}\frac{dx}{x} = \frac{\mu_0\,\omega\lambda}{4\pi}(\ln 2)$$

To find the magnetic moment,

$$M = \int (di\,A) = \int dq\left(\frac{w}{2\pi}\right)\left(\pi x^2\right)$$

$$= \left(\frac{\lambda\omega}{2}\right)\int_l^{2l} x^2\, dx = \frac{\lambda\omega}{6}(7l^3)$$

$$M = \left(\frac{7Qwl^2}{6}\right)$$

Example 21

A non conducting spherical shell of radius R with uniform surface charge density σ, is rotated with constant angular velocity ω about its diameterical axis. Then find the magnetic induction at the centre.

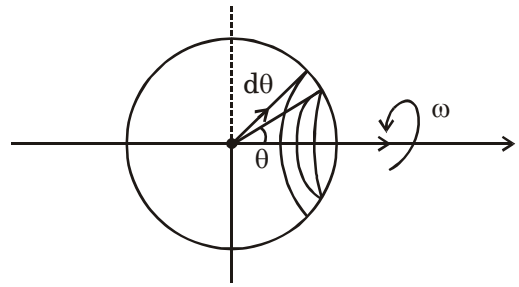
Solution :

Consider an elemental ring of thickness Rdθ making an angle θ with the horizontal. Now, charge on it.

$$dq = \sigma\,(2\pi R\sin\theta)\,Rd\theta$$

$$di = \left(\frac{dqw}{2\pi}\right)$$

$$= \int \frac{\mu_0 \ di R^2 \sin^2\theta}{2(R)^3}$$



$$B = \frac{\mu_0 \ \omega}{2R(2\pi)} \sigma (2\pi R^2) \int_0^{\pi} \sin^3\theta \, d\theta$$

$$= \frac{\mu_0 \ \omega\sigma R}{2} \times \frac{4}{3}$$

$$B = \left(\frac{2\mu_0 \ w\sigma R}{3}\right)$$

To calculate the magnetic moment,

$$\frac{M}{L} = \left(\frac{q}{2m}\right) \qquad \therefore \ M = \frac{q}{2m}\left(\frac{2}{3} mR^2\right)\omega$$

$$\boxed{M = \left(\frac{qR^2}{3}\right)\omega}$$

Example 22

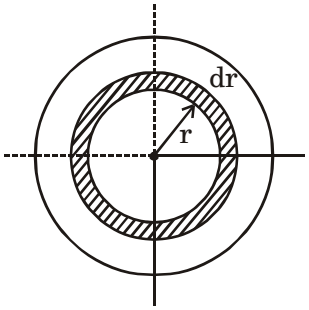
In the previous example, find the magnetic induction at the centre if the sphere is solid.

Solution :

In the previous questions we have already found the magnetic field due to the thin spherical shell

$$B = \left(\frac{2}{3}\mu_0 \ \omega\sigma R\right).$$

In this case, consider this spherical concentric shells of thickness dr at a distance r from the centre of the sphere.



$$dB = \left(\frac{2}{3}\mu_0 \omega \ \sigma \ r\right)$$

Use $\boxed{\sigma = \rho dr}$

$$\therefore\ dB = \frac{2}{3}\mu_0\ w(\rho dr)r = \frac{2}{3}\mu_0\ w\rho r dr$$

$$B = \int dB = \frac{2}{3}\mu_0 \rho w \int_0^R r dr$$

$$= \frac{2}{3}\ \frac{\mu_0 \rho \omega R^2}{2}$$

$$\therefore \quad B = \left(\frac{\mu_0 \rho w R^2}{3}\right)$$

Properties of Magnetic Lines of Forces :

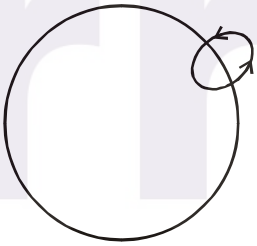
(1) Magnetic lines of forces always form a closed loop.

(2) Magnetic lines of force may or may not be terminating at right angle to the surface of the magnetic material.

(3) Magnetic lines of forces originate from the north pole and end at the south pole.

(4) The magnetic flux through a closed surface is always zero.

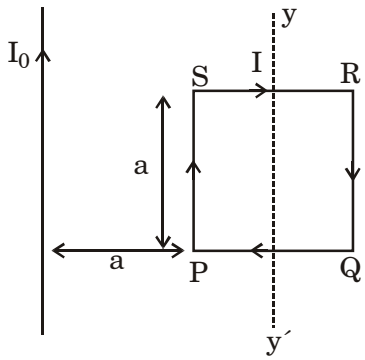
Reason : Since the magnetic lines form a closed loop, the number of field lines coming out and join into the surface are same. Hence net flux = 0.



$$\oint \vec{B} \cdot \vec{dr} = 0$$

Example 23

A square current loop of side a is located in the same plane as that of a long current wire, carrying a current I_0. Find the net ampere force acting on the loop.



Solution :

To find the force acting on the loop

(Note that the force in the arms PQ and RS are equal and opp. and hence they cancel each other).

Force on arm PS

$$= (IaB) \quad = Ia\left(\frac{\mu_0 I_0}{2\pi a}\right) = \frac{\mu_0 I I_0}{2\pi} \text{ (towards the longwire)}$$

$$\text{Force on the arm QR} = IaB = Ia\left(\frac{\mu_0 I_0}{2\pi 2a}\right) = \left(\frac{\mu_0 I I_0}{4\pi}\right) \text{ (away from long wire)}$$

$$\therefore \text{ Net force on the loop} = \frac{\mu_0 I I_0}{2\pi} - \frac{\mu_0 I I_0}{4\pi} = \frac{\mu_0 I I_0}{4\pi}$$

Example 24

Two long conductors are placed in the same plane at a separation 'a' as shown. Find the force of interaction per unit length between the two conductors.

Solution :

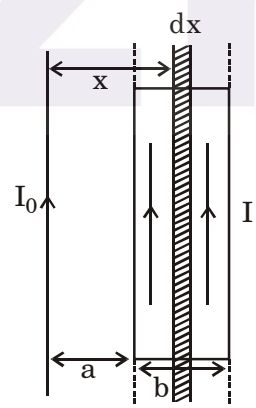
Consider a strip at a distance x from the long wire of the thick wire, which again behaves as if a long wire.

$$di = \left(\frac{I}{b}\right)dx$$

$$\text{Force per unit length} = \left(\frac{\mu_0 i_1 i_2}{2\pi x}\right)$$

$$\therefore \quad F = \int_a^{(a+b)} \frac{\mu_0 I_0 dr}{2\pi x} = \int_a^{(a+b)} \frac{\mu_0 I_0}{2\pi}\left(\frac{I}{b}\right)\frac{dx}{x}$$

$$\therefore \quad F = \left(\frac{\mu_0 I_0 I}{2\pi b}\right)\left[\ln\frac{(a+b)}{a}\right]$$



Example 25

A circular current loop is placed at a distance 2R from a long straight conducting wire as shown in the same plane. Find the force of interaction between them.

Solution :

Consider an arc element $d\theta$ at an angular position of θ.

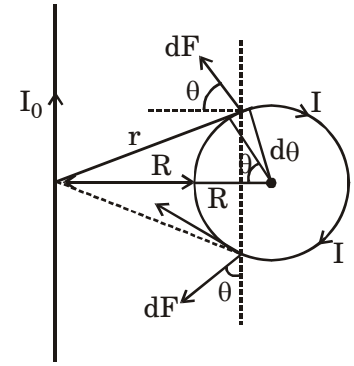
Net force will be acting along the x-axis

$\therefore \int dF \sin\theta = 0$

$\therefore \quad F_{net} = \int dF \cos\theta$

$$= \int \frac{(IRd\theta)\ \mu_0 I_0 \cos\theta}{2\pi\ (2R - R\cos\theta)}$$

$$\therefore \quad F_{net} = \frac{\mu_0 I_0 IR}{2\pi R} \int_0^{2\pi} \frac{\cos\theta\, d\theta}{2-\cos\theta}$$



Example 26

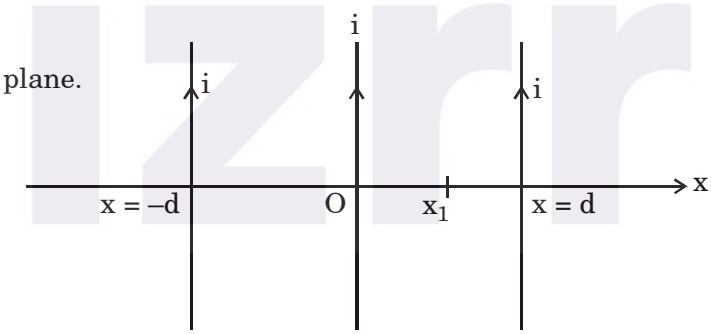
Three long straight wires each carrying current (i) are located in X – Y plane. The middle wire is placed at x = 0, while other two conductors are placed at x = ± d.

(i) **Find the locus of points where net $\vec{B}$ is zero.**

(ii) **If the central wire is slightly displaced along the Z-axis and then released. Then find the time period of small oscillations.**

Solution :

Note that $\vec{B}$ field will be zero only in xy plane.



(i) $\vec{B}$ field will be zero at x = 0

At any other point along x axis

$$\frac{\mu_0 i}{2\pi x_1} + \frac{\mu_0 i}{2\pi(x_1 + d)} = \frac{\mu_0 i}{2\pi(d - x_1)}$$

$$\frac{1}{x} + \frac{1}{(d+x)} = \frac{1}{(d-x)}$$

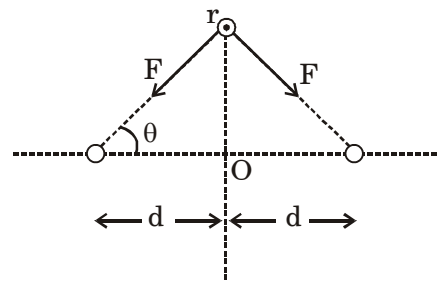
$$\frac{1}{x} = \frac{1}{(d-x)} - \frac{1}{(d+x)} = \frac{2x}{d^2 - x^2}$$

$d^2 - x^2 = 2x^2$

$d^2 = 3x^2$

$\therefore \boxed{x = \pm\left(\frac{d}{\sqrt{3}}\right)}$ required locus with (x = 0)

(ii) Net force = 2F sinθ (towards origin)



$$\lambda\left(\frac{d^2x}{dt^2}\right) = -2F\sin\theta$$

$$\Rightarrow \lambda\left(\frac{d^2x}{dt^2}\right) = -\frac{2\mu_0 i^2}{(2\pi r)}\sin\theta \qquad = -\frac{\mu_0 i^2 x}{\pi\left[(d^2+x^2)^{1/2}\right]^2}$$

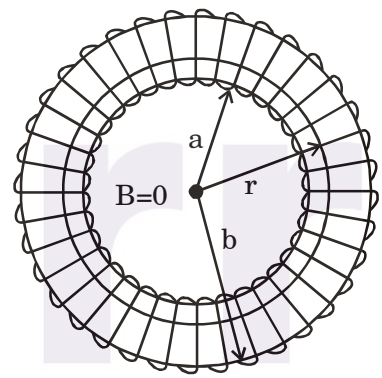
$$\left(\frac{d^2x}{dt^2}\right) = \frac{-\mu_0 i^2 x}{\pi\lambda d^2} \text{ if } (d >> x)$$

$$\therefore \omega = \sqrt{\frac{\mu_0 i^2}{\pi\lambda d^2}} \qquad \therefore T = 2\pi\sqrt{\frac{\pi\lambda d^2}{\mu_0 i^2}}$$

10. TOROID

Current in the toroid = i

Total no. of turns = N



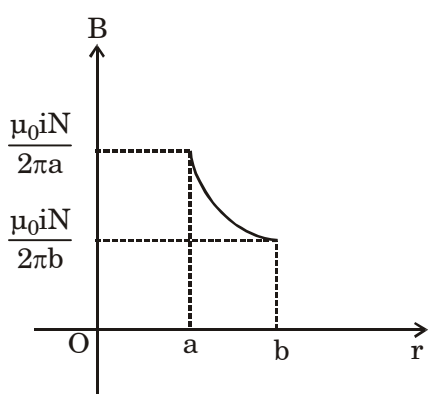
$$\oint \vec{B}\cdot\vec{dl} = \mu_0 I_{net}$$

$$B(2\pi r) = \mu_0 (iN)$$

$$\boxed{\therefore B = \frac{\mu_0 i N}{2\pi r}}$$

Thus, B = 0 for (r < a) (r > b)

$$B = \left(\frac{\mu_0 i N}{2\pi r}\right) \text{ for } (a \le r \le b)$$



$$\left(B \,\alpha\, \frac{1}{r}\right)$$

However, if r is very large compared with the cross sectional radius of the toroid, the field is approximately uniform.

$$\frac{N}{2\pi r} = n = \text{number of turns / length}$$

$$\boxed{\therefore B = \mu_0 ni}$$

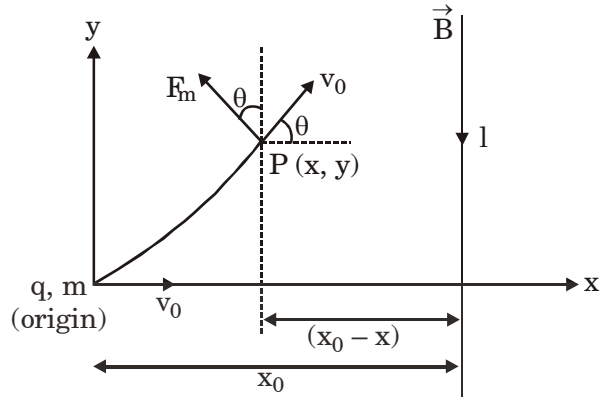
MISCELLANEOUS EXAMPLES

Example 1

A long straight wire carries a current i. **A particle having a positive charge** q **and mass** m**, kept at a distance** x_0 **from the wire is projected towards it with a speed** v_0**. Find the minimum separation between the wire and the particle.**

Solution :

Magnetic field at point P due to current i is,



$$B = \frac{\mu_0}{2\pi}\frac{i}{(x_0 - x)} \quad \text{in } \otimes \text{ direction.}$$

Now, $a_y = \dfrac{F_m}{m}\cos\theta$

or $$\frac{dv_y}{dt} = \frac{\left(\dfrac{\mu_0}{2\pi}\dfrac{i}{x_0 - x}\right)(qv_0\cos\theta)}{m}$$

$$\therefore \quad \frac{dv_y}{dx}\cdot\frac{dx}{dt} = \left(\frac{\mu_0 iq}{2\pi m}\right)\left(\frac{1}{x_0 - x}\right)(v_0\cos\theta)$$

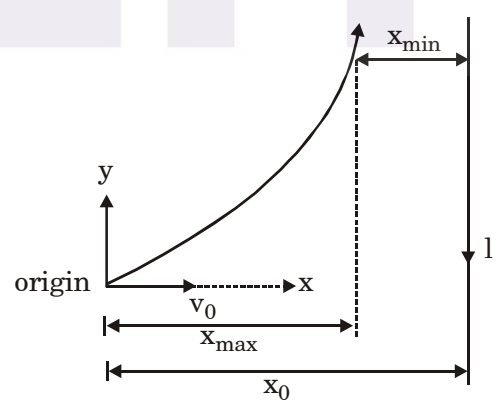
Here $\dfrac{dx}{dt} = V_x = v_0\cos\theta$

$$\therefore \quad \frac{dv_y}{dt} = \left(\frac{\mu_0 qi}{2\pi m}\right)\left(\frac{1}{x_0 - x}\right)$$

$$\therefore \quad \int_0^{v_0} dv_y = \left(\frac{\mu_0 qi}{2\pi m}\right)\int_0^{x_{max}}\frac{dx}{x_0 - x}$$

$$\therefore \quad \int_0^{v_0} dv_y = \left(\frac{\mu_0 qi}{2\pi m}\right)\int_0^{x_{max}}\frac{dx}{x_0 - x}$$

$$\therefore \quad v_0 = -\left(\frac{\mu_0 qi}{2\pi m}\right)\ln\left(\frac{x_0 - x_{max}}{x_0}\right)$$



$$\therefore \quad x_0 - x_{max} = x_0\, e^{\frac{-2\pi m v_0}{\mu_0 qi}}$$

Here, $x_0 - x_{max}$ is the minimum distance between the particle and the wire.

$$\therefore \quad x_{min} = x_0\, e^{\frac{-2\pi m v_0}{\mu_0 qi}} \quad \textbf{Ans.}$$

Example 2

Two long straight conductors are placed at y = ± a along the z-axis and carry current I. A current carrying conductor AC is placed along x-axis.

(i) **Find the net force experienced by the current segment AC.**

(ii) **Also, find the force when the current in one wire is reversed.**

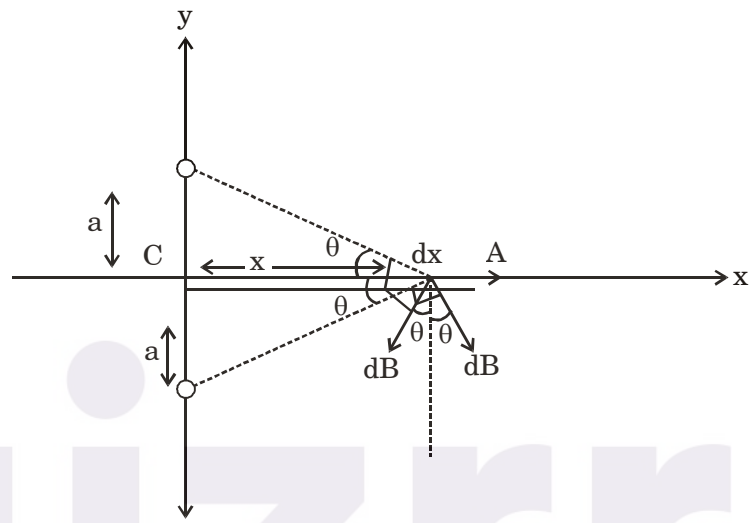
Solution :

Consider an element of length dx at a distance x from origin.

Now, net $\vec{B}$ acting on the element (dx)

$$= 2d\ B \cos\theta$$

$$= 2\left(\frac{\mu_0 I_0}{2\pi r}\right)\left(\frac{x}{r}\right) = \frac{\mu_0 I_0 (x)}{\pi (a^2 + x^2)}$$



$$\therefore \quad dF = (dB)_{net}\left(\frac{I}{l}\right)dx$$

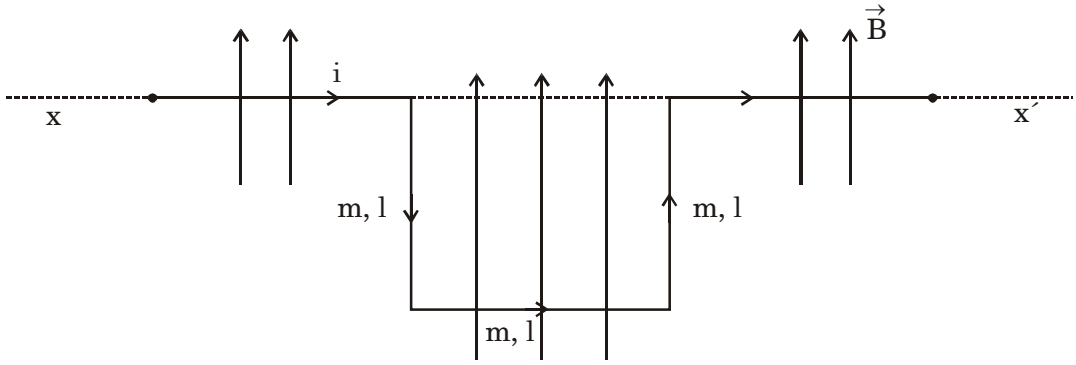
$$= \frac{\mu_0 I_0 x}{\pi (a^2 + x^2)}\ dx\left(\frac{I}{l}\right)$$

$$\therefore \quad F = \frac{\mu_0 I_0 I}{\pi l}\int_0^l \frac{x\,dx}{(d^2 + x^2)}$$

$$= \frac{\mu_0 I_0 I}{\pi l}\left[\frac{1}{d} - \frac{1}{\sqrt{d^2 + l^2}}\right]$$

Example 3

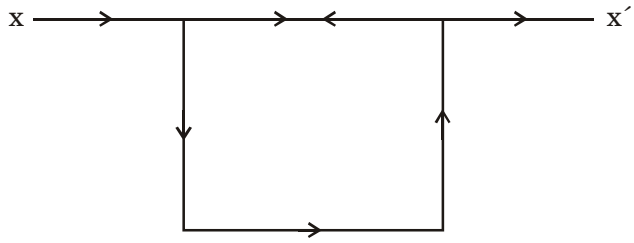
A wire of mass 3 m and length 3l is bent in the shape of 3 sides of a square. It can freely rotate about the axis XX′ about a vertical plane.



A uniform magnetic field is switched on in the region and a current i is passed through the wire. Then, find :

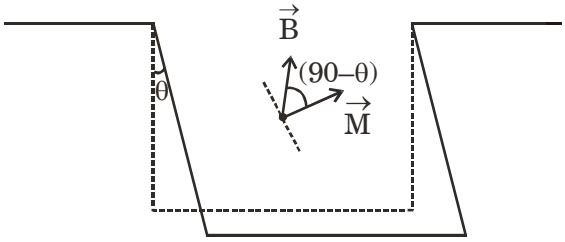
(i) the deflection of the frame from the vertical plane in equilibrium.

Solution :



To complete the wire frame loop, replace the incomplete side with current wires having current as shown. The torque of the single wire about the axis XX´ is zero. {Here we are taking about the torque of the wire formed due to super position method}.

Torque due to the magnetic field



$$\vec{\tau} = \vec{M} \times \vec{B}$$

$$= MB \sin(90 - \theta) = MB \cos\theta$$

$$= B_i (l^2) \cos\theta.$$

{This is torque when the loop has rotated by an angle θ}

This torque is balanced by the torque due to the weight.

$$\tau_{wt} = \left(mg\frac{l}{2}\sin\theta\right) + (mgl\sin\theta) + \left(mg\frac{l}{2}\sin\theta\right)$$

$$= 2\ mg\ l \sin\theta$$

Equate these at the equilibrium position,

$Bil^2 \cos = 2mgl \sin\theta$

$$\therefore \quad \tan\theta = \frac{Bil}{2mg}$$

$$\therefore \quad \boxed{\theta = \tan^{-1}\left(\frac{Bil}{2\ mg}\right)}$$

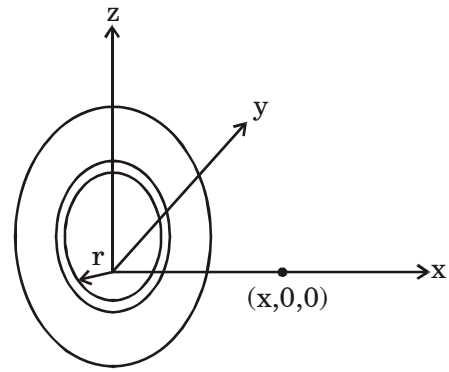
Example 4

A charge Q is uniformly distributed over the surface of a non-conducting circular disc. It is now rotated with constant angular velocity w, about an axis perpendicular to the plane of the disc passing through its centre.

Then find the magnetic induction at a distance x from the centre on the axis.

Solution :

Consider a circular element at distance r from the centre of the disc.



$$\text{Surface charge density} = \frac{\sigma}{\pi R^2}$$

$$\therefore \quad dq = \sigma\,(2\pi r)\,dr$$

$$\therefore \quad di = (dq)\left(\frac{w}{2\pi}\right) = \frac{Qw\,(2\pi r)\,dr}{2\pi^2 R^2}$$

$$= \frac{Qw(rdr)}{\pi R^2}$$

Hence, $$dB = \frac{\mu_0\, di\, r^2}{2(x^2 + r^2)^{3/2}}$$ {This is due to an elemental disc}

$$= \frac{\mu_0\, Qw}{2\pi R^2} \int_0^R \frac{r^3 dr}{(r^2 + x^2)^{3/2}}$$

$$\boxed{B_{net} = \frac{\mu_0\, Qw}{2\pi R^2}\left\{\frac{R^2 + 2x^2}{\sqrt{R^2 + x^2}} - 2x\right\}}$$

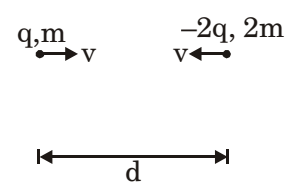
Example 5

A charged particle +q of mass m is placed at a distance d from another charged particle – 2q of mass 2 m in a uniform magnetic field B as shown in figure. If the particles are projected towards each other with equal speeds v.

(a) **Find the maximum value of projection speed v_m so that the two particles do not collide.**

(b) **Find the time after which collision occurs between the particles if projection speed equals $2v_m$.**

(c) **Assuming the collision to be perfectly inelastic find the radius of particle in subsequent motion. (Neglect the electric force between the charges).**



Solution :

(a) The particles will move in circular paths, as velocity vector is perpendicular to magnetic field. Time period of both the particles is same $\left(T = \frac{2\pi m}{Bq}\right)$. So, for collision not to take place,

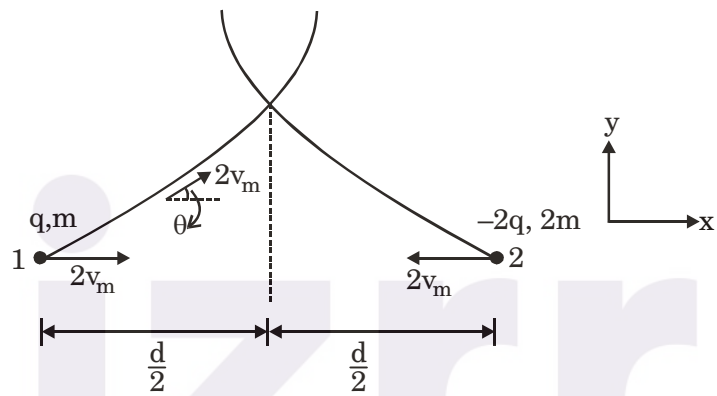
$$r_1 + r_2 < d$$

or $$\frac{mv}{Bq} + \frac{2mv}{2Bq} < d \quad \text{or} \quad v < \frac{Bqd}{2m}$$

Therefore, maximum speed should be

$$\frac{Bqd}{2m} \quad \text{or} \quad v_m - \frac{Bqd}{2m}$$

(b) From symmetry they collide at $\frac{d}{2}$.



$$\theta = \omega t = \left(\frac{Bq}{m}\right) t$$

x-component of velocity at time t is,

$$v_x = 2v_m \cos\theta = 2v_m \cos\left(\frac{Bq}{m}\right) t$$

$$\therefore \quad \frac{d}{2} = \int_0^t v_x dt = 2v_m \int_0^t \cos\left(\frac{Bq}{m} t\right) dt$$

or $$\frac{d}{2} = \frac{2mv_m}{Bq}\left[\sin\left(\frac{Bq}{m} t\right)\right]_0^t$$

$$\frac{d}{2} = \frac{2mv_m}{Bq} \sin\left(\frac{Bq}{m} t\right)$$

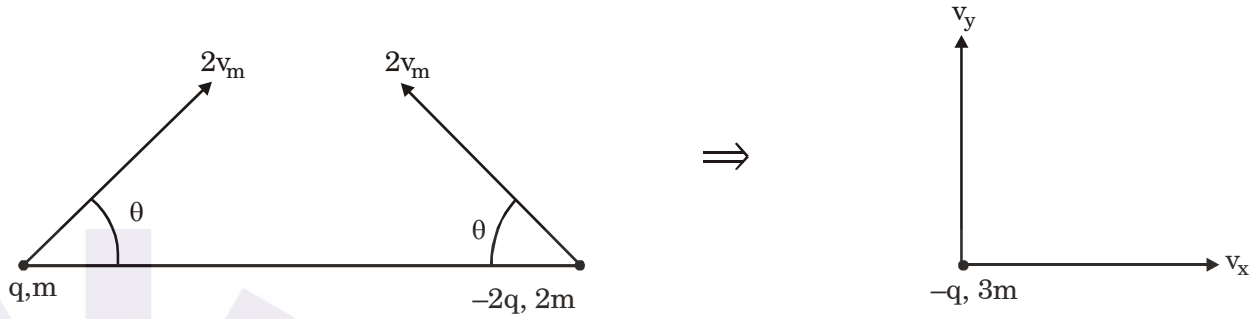
Substituting $v_m = \frac{Bqd}{2m}$, we have

$$\sin\left(\frac{Bq}{m} t\right) = \frac{1}{2}$$

$\therefore \qquad \dfrac{Bqt}{m} = \dfrac{\pi}{6}$

or $\qquad t = \dfrac{m\pi}{6Bq}$ **Ans.**

(c) At the time of collision, $\theta = \omega t = \left(\dfrac{Bq}{m}\right)\left(\dfrac{m\pi}{6Bq}\right) = \dfrac{\pi}{6} = 30°$



After the collision :

$$v_x = \frac{2mv_m \cos 30° - 4mv_m \cos 30°}{3m} = \frac{v_m}{\sqrt{3}}$$

$$v_y = \frac{2mv_m \sin 30° + 4mv_m \sin 30°}{3m} = v_m$$

$\therefore$ Speed of combined mass $v = \sqrt{v_x^2 + v_y^2} = \dfrac{2}{\sqrt{3}} v_m$

$$\therefore \quad R = \frac{(3m)(v)}{Bq} = \frac{(3m)\left(\frac{2}{\sqrt{3}} v_m\right)}{Bq} = \frac{2\sqrt{3}\, mv_m}{Bq}$$

Substituting, $v_m = \dfrac{Bqd}{2m}$, we get $\qquad R = \sqrt{3}d$ **Ans.**